

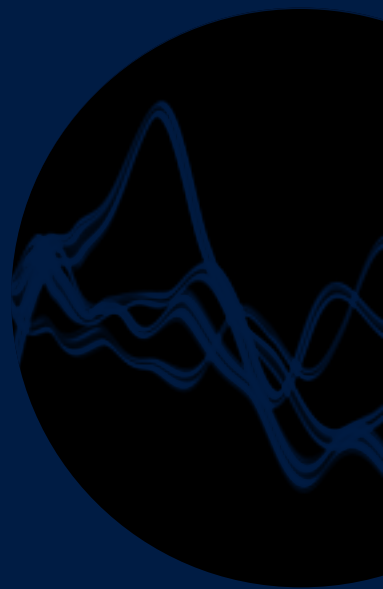
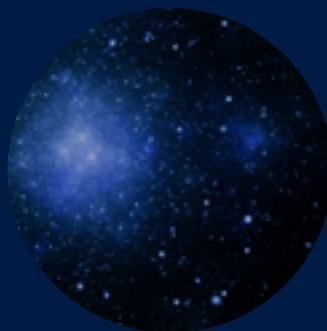
Review

ERICSSON
TECHNOLOGY

ARTIFICIAL
INTELLIGENCE
IN RAN

VERSATILE
VIDEO CODING
EXPLAINED

APPLIED
NETWORK SLICING
SCENARIOS







08 MONITORING IoT APPLICATION PERFORMANCE WITH MACHINE QoE

Awareness of the QoE that the network delivers is essential for all mission-critical Internet of Things (IoT) systems. To reliably predict QoE in IoT applications, we have developed a generic Machine QoE framework and tested a vertical-specific version of it in a smart-grid scenario.



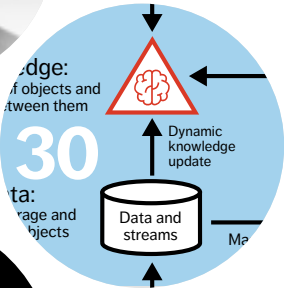
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Versatile Video Coding – the new video compression coding standard that was recently approved by MPEG and the ITU-T – represents state-of-the-art video coding and is certain to play an important role in supporting a wide range of 5G use cases.



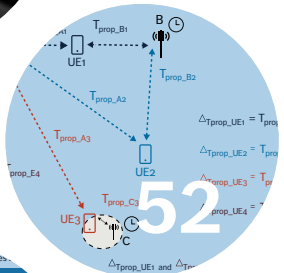
30 COGNITIVE PROCESSES FOR ADAPTIVE INTENT-BASED NETWORKING

Extensive automation will be necessary to cope with the unprecedented flexibility and dynamic adaptation that 5G networks are introducing into service delivery and network resource utilization. Our intent-based networking approach combines several artificial intelligence (AI) techniques within a flexible architecture and uses intents to specify what the autonomous system is expected to do.



42 ARTIFICIAL INTELLIGENCE (AI) IN RAN – A SOFTWARE FRAMEWORK FOR AI-DRIVEN RAN AUTOMATION

When used correctly, AI and machine learning techniques have tremendous potential to help overcome complex cross-domain automation challenges in radio networks. Ericsson has identified the key software enablers for AI-based RAN automation and integrated them into a comprehensive framework that provides a solid and flexible technological foundation.

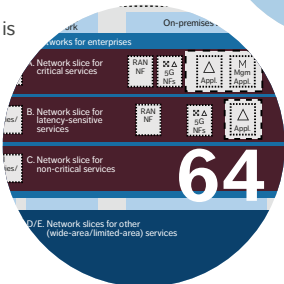


52 5G SYNCHRONIZATION REQUIREMENTS AND SOLUTIONS

Achieving proper network synchronization in 5G requires more planning than it did in previous generations due to the fact that there is no single synchronization requirement in 5G. A solid understanding of 5G timing needs and their associated solutions is essential for an efficient and cost-effective implementation.

64 APPLIED NETWORK SLICING SCENARIOS IN 5G

Many new and emerging 5G use cases require network slicing, an outcome that is achieved through the use of an evolving toolbox of enablers in five areas: cloud infrastructure, RAN, core, transport and operations support systems/business support systems. Depending on the scenario, different combinations of enablers are needed to engineer the appropriate network slice(s).



Ericsson Technology Review brings you insights into some of the key emerging innovations that are shaping the future of ICT. Our aim is to encourage an open discussion about the potential, practicalities, and benefits of a wide range of technical developments, and provide insight into what the future has to offer.

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Creating intelligent digital infrastructure
by Erik Ekudden

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BUILDING BACK BETTER ON THE NETWORK PLATFORM

■ **THE UNPRECEDENTED** events of 2020 have provided us all with ample evidence that the technology and solutions Ericsson delivers in partnership with our customers are fundamental to the functioning and future development of both industry and society as a whole.

As governments and enterprises prepare to “build back better” post-pandemic, we at Ericsson are excited about using our technology platform to help communication service providers (CSPs) develop innovative solutions that are smarter, faster and more efficient than ever before. The ability to leverage artificial intelligence (AI) techniques and automate as much as possible will be key to our mutual success.

While telecom AI is still very much a work in progress, our research indicates that it is already possible to reach a high degree of practical autonomous operation in networks by combining existing AI techniques within a flexible architecture to form what we at Ericsson call a cognitive layer. The article on page 30 highlights the critical role intents play in creating this new layer and enabling autonomous operations.

Building the next generation of autonomous systems will also require AI-powered automation of the RAN. The article on page 42 presents a comprehensive framework of the key software enablers of AI-based RAN automation, along with two proof-of-concept examples executed in real RAN contexts.

...SOLUTIONS THAT ARE SMARTER, FASTER AND MORE EFFICIENT THAN EVER BEFORE

While autonomous operations are going to be of great value in a broad range of different applications and scenarios, the industrial automation use case is definitely one of the most rigorous due to its stringent timing requirements. The 5G synchronization article on page 52 explains the main categories of 5G synchronization requirements and presents a toolbox of solutions for all relevant scenarios, the most challenging of which can often be addressed within the RAN domain.

Private 5G networks created with the help of network slicing techniques are likely to become very popular with both private enterprises and public agencies in the years ahead. The article on page 64 highlights the benefits of these types of networks and explains how CSPs can use an evolving set of enablers in cloud infrastructure, the RAN, the core and transport networks, and OSS/BSS to create appropriate slices for each use case.

The final two articles in this issue of the magazine explore recent breakthroughs in video compression technology and take a closer look at the requirements of mission-critical Internet of Things

(IoT) systems in terms of QoE. To ensure a consistently high level of customer satisfaction in these types of systems, we have created a Machine QoE framework that is specifically designed to measure the QoE of IoT devices.

We hope you enjoy this issue of our magazine and we'd be delighted if you shared it with your colleagues and business partners. You can find both PDF and HTML versions of all the articles at: www.ericsson.com/ericsson-technology-review



Erik Ekudden

ERIK EKUDDEN

SENIOR VICE PRESIDENT,
CHIEF TECHNOLOGY OFFICER AND
HEAD OF GROUP FUNCTION TECHNOLOGY

Monitoring IoT application performance

WITH MACHINE QoE

New capabilities in 5G make it ideal to provide cost-effective solutions for mission-critical Internet of Things systems that depend on ultra-reliability and low latency. A necessary component when creating these solutions is an awareness of the QoE that the network offers the IoT system.

CONSTANT WETTE
TCHOUATI,
STEVEN ROCHEFORT,
GEORGE SARMONIKAS

The Internet of Things (IoT) is a worldwide network of physical objects – buildings, cars, wearables, industrial machines and so on – that are equipped with connectivity devices to build a communication network where connected objects can exchange data with other objects.

■ Almost anything can be connected to anything in the IoT, creating a complex network of connected machines, from simple wireless tags to sensors or actuators with capabilities to sense, communicate, process data or control their environment. The potential to apply IoT technology is endless, stretching across all industry verticals, resulting in a wide variety of devices and communications requirements on the underlying network. Large-scale deployment of cellular IoT devices is expected in the years ahead with the widespread introduction of 5G technology.

For communications service providers (CSPs), a typical IoT subscriber is an enterprise or a service provider from any vertical that wants to enable connectivity in many devices spread in one region or worldwide. Given the specificity and variability of IoT requirements per vertical, CSPs need a new approach in the assessment of IoT subscribers' QoE. To meet this need, Ericsson has developed a generic Machine QoE (M-QoE) framework for accurate prediction of IoT subscribers' QoE and tested a vertical-specific version of it in a smart-grid scenario.

Creating a framework for Machine QoE

IoT networks support multiple technologies, standards, device types and application types with various performance requirements between verticals and between applications of the same vertical. A preliminary step to design an M-QoE model for this complex environment is to create classes of IoT applications of the same vertical

or similar performance requirements. The model features can then be optimized for each CSP according to the specific requirements of their network environment.

Classification of IoT applications

There are several ways to classify IoT applications. One approach is to classify them according to business verticals such as health care, agriculture, automotive, transportation, surveillance, smart home, smart city, smart grid and smart metering. Each vertical comprises applications with various degrees of performance requirements ranging from very low to high data speed, from best effort to ultra-reliable, and from no latency to ultra-low latency.

From a network perspective, applications are often classified into three categories based on their network performance requirements: massive IoT, mission-critical control and enhanced mobile broadband (eMBB) [1]. Massive IoT requirements include deep coverage, ultra-low device energy consumption, ultra-low complexity and ultra-high density. Mission-critical control requirements include strong security, ultra-high reliability, ultra-low latency and extreme user mobility. The eMBB requirements include extreme capacity, extreme data rates and deep awareness (for discovery and optimization).

The 3GPP specifications define three main categories of applications, two of which are dedicated to IoT, namely ultra-reliable and low-latency communications (URLLC) and massive Machine-Type Communications (mMTC).

URLLC provides the ultra-reliability or ultra-low latency required for numerous use cases in smart grid, intelligent transportation systems and smart manufacturing, while mMTC enables connectivity to lots of low-power and low-cost devices for best-effort applications and long-range coverage in smart cities or smart logistics.

At Ericsson, we have defined four categories of IoT applications: Massive IoT, Critical IoT, Broadband IoT and Industrial Automation IoT. The first two correspond to the 3GPP mMTC and URLLC categories respectively. Broadband IoT provides the high-data rates and large data volume required for unmanned aerial vehicles/drones and augmented reality/virtual reality applications. Industrial IoT provides the precise indoor positioning and time-sensitive networking for real-time applications that require deterministic communication.

The M-QoE framework that we have designed generalizes on the four IoT categories by inferring from specific cases. For this purpose, we first investigated the important characteristics of IoT applications in three business verticals: utilities, automotive and smart cities [2].

KPIs for machines

Our framework to assess the QoE of IoT subscribers comprises a list of common IoT KPIs as shown in *Table 1*. These KPIs are calculated on a per device, per subscriber and per vertical basis. The threshold values are based on the application type and the IoT vertical that the device belongs to.

Terms and abbreviations

AI – Artificial Intelligence | CSP – Communications Service Provider | E-KPIs – Expected Value KPIs | eMBB – Enhanced Mobile Broadband | IoT – Internet of Things | KPI – Key Performance Indicator | M-QoE – Machine QoE | ML – Machine Learning | mMTC – Massive Machine-Type Communications | O-KPIs – Observed KPIs | PDC – Phasor Data Concentrator | PMU – Phasor Measurement Unit | URLLC – Ultra-Reliable and Low-Latency Communications | WAMS – Wide Area Measurement System

KPI	Definition
latency (millisecond)	the time it takes to transfer a given piece of information from the moment it is transmitted by the source to the moment it is successfully received at the destination
packet loss (ratio)	the percentage of frames that should have been forwarded by a network but were not
bit error ratio	the number of bit errors divided by the total number of transferred bits during a studied time interval
energy efficiency (joule/byte)	the energy consumed for the end-to-end transport of a byte
security	level of importance for attack prevention
data (messages) rate, downlink/uplink (bit/s)	a time-variable function that might be important to define some parameters (such as peak, burst, average, minimum, maximum) in order to better describe the data rate (messages)
jitter (ms)	the short-term variations of a digital signal's significant instants from their ideal positions in time
packet delay variation (ms)	variation in latency as measured in the variability over time of the packet latency across a network (expressed as an average of the deviation from the network mean latency)
reliability	percentage of sent network layer packets successfully delivered to a given node within the time constraint required by the targeted service, divided by the total number of sent network layer packets
availability	the percentage of available time (with respect to total time) in a generic observation period of the connection across the transport network
mobility (km/h)	fixed (no mobility: office, home) or maximum speed in movement (pedestrian or using a means of transportation such as a train, road vehicle, airplane or drone)
traffic density (Mbit/s/sq km)	traffic in a specific area
connection density (devices/sq km)	number of devices in a specific area
coverage (sq km)	area of application interest
battery lifetime (days, years)	time of battery duration
data size (bytes)	size of the atomic packet or frame (average, maximum)
location accuracy	the maximum positioning error tolerated by the application (can be measured outdoor and indoor in 5G)
cost	overall cost including connection, device and cloud services

Table 1 List of common IoT KPIs

Framework architecture for Machine QoE

The system architecture for the M-QoE framework has the same common base as Ericsson’s operational support system for subscriber service assurance. It is the foundation to provide not only consistent IoT service assurance but also an enhanced experience for the users of the connected machines (IoT devices) across several touchpoints along the use case journey.

Figure 1 shows the high-level, end-to-end architecture for our M-QoE framework. Machines of different kinds are wirelessly connected over 5G (or other cellular technologies). The yellow dots indicate the points where the assurance metrics related to M-QoE are usually measured:

- » the IoT device
- » the RAN
- » the edge network
- » the core network/services network
- » vertical slices and service layers (not shown).

The metrics are then ingested and processed at different points of the network subject to the processing delay requirements of each use case.

There can be a combination of RAN, edge and centralized processing of the received metrics. Stream processing at the edge is enabled when low latency and real-time processing are required, whereas processing takes place at the central data center in cases where there is no low-latency requirement. A great variety of interfaces allow for raw data to be preprocessed prior to the generation of any insights. Processed data is stored in a distributed database to make it available for further analysis.

A rules engine and machine learning (ML) models are used to provide M-QoE scores along with other insights such as anomalies, patterns of use, mobility patterns and other use-case-specific insights. The cognitive reasoning system, driven by business intents such as the proactive service level assurance of smart-grid infrastructure, can use the insights for autonomic decision-making. It can continuously maintain autonomously the desired service level specification by recommending actions to the service/operations engineers and/or triggering closed loop actions with minimal human intervention.

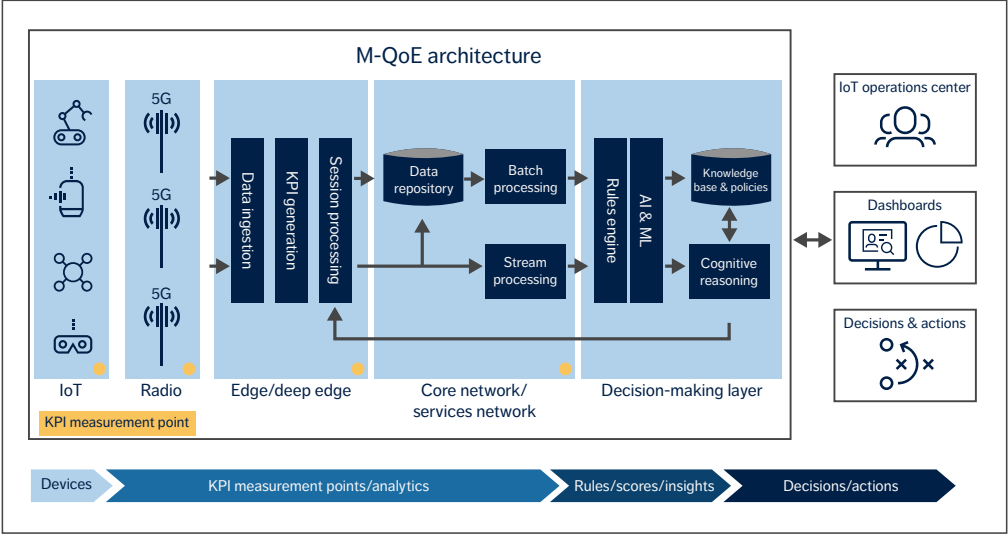


Figure 1 M-QoE framework architecture

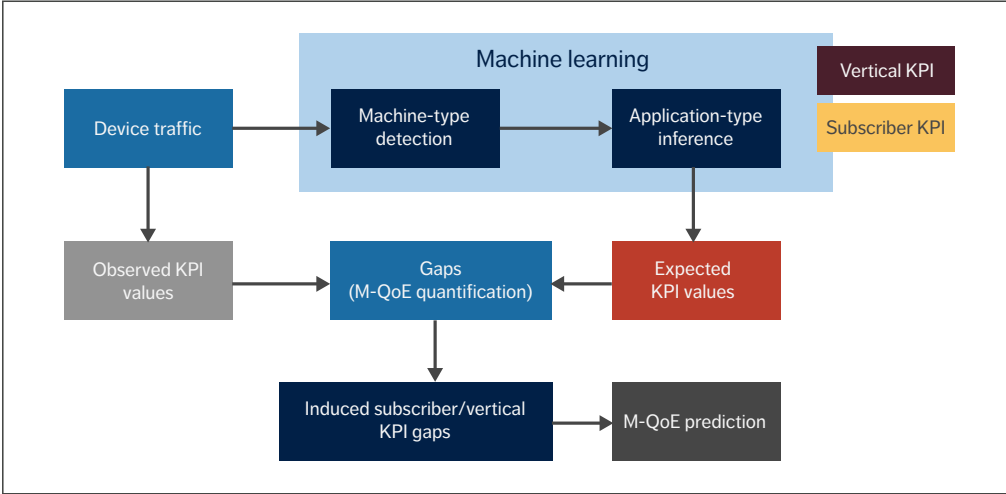


Figure 2 The architecture of ML models in M-QoE

All generated insights, recommendations and actions are exposed through dashboards to various business and operational level management systems including IoT service operations, and usually used for planning, management and decision-making purposes [3].

Machine QoE models

To improve the performance of our M-QoE framework, we added some rule-based and ML models to detect insights or anomalies in KPI measurements that operators could use to improve IoT subscribers’ network experience, as shown in Figure 2.

Our M-QoE framework includes four models:

- » machine-type detection
- » application-type inference
- » M-QoE quantification
- » M-QoE prediction.

CSPs can use the machine detection model in our M-QoE framework to detect if a device accessing their network is an IoT device or a mobile phone to better manage their QoE. This is helpful because, unlike mobile phones, there is no global database of IoT devices, and the IoT landscape is highly

fragmented with many device models and service providers.

Our M-QoE framework also includes an ML model for IoT segmentation, which was trained with samples of traffic data of known IoT applications. This model can be used in production to detect traffic patterns and predict the type of IoT application and vertical. This is valuable, as the IoT ecosystem is fueled by a multitude of small and medium vertical service providers whose traffic patterns have yet to be discovered.

The M-QoE quantification in our framework is done by computing the gap between the observed KPIs (O-KPIs) and their expected values (E-KPIs) defined by the Service Level Agreement. It should be noted that a typical IoT subscriber owns multiple devices, and their overall satisfaction with a service is a function of multiple features that are monitored by the different O-KPIs. To compute the overall M-QoE of the IoT subscriber, the first step is to aggregate the O-KPI measurements of all their devices and derive a feature-specific M-QoE_i for feature i. The second step is to add up the feature-specific M-QoE_i weighted by their importance factor to obtain the overall M-QoE.

The computation of an IoT subscriber’s O-KPI is done by aggregating the measurements of their devices plus measurements at the vertical node over different dimensions: temporal, spatial, devices and services. Our aggregation procedure consists of computing the following statistics: average, variance, skewness, kurtosis, percentile (5, 25, 50, 75 and 95), minimum, maximum and range. The M-QoE_i score is derived from the dataset of the difference (O-KPI_i - E-KPI_i) between the two measurements and using an ML algorithm on a scale of 1 to 5 labelled as (1) bad, (2) poor, (3) fair, (4) good or (5) excellent.

The overall M-QoE value is $M-QoE = \sum [\alpha_i * M-QoE_i]$ where α_i represents the importance level of feature i. Mobile operators can optimize the performance of this M-QoE model by selecting a subset of KPIs that are relevant to their network environment.

The purpose of the M-QoE prediction model is to overcome the challenge of CSPs not being able to probe the IoT devices or the vertical’s network infrastructure to access measurements of subscriber KPIs and vertical KPIs, as these are external environments. These measurements are usually obtained from the MOS (mean opinion score) test, but this is done only once, limiting the dynamic tracking of user satisfaction. To address these limitations, we trained an ML algorithm to classify and predict the unknown E-KPI measurements from the known O-KPI factors records. The M-QoE prediction model uses an inductive supervised learning approach.

Case study: Machine QoE in next-generation power grids

Electric power grids are one of the verticals that will benefit most from M-QoE in the near term. The business model in this vertical is undergoing a transformation from mega grids operated by one large company to segmented grids and micro grids, giving consumers the possibility to choose between several offerings from different suppliers. In this competitive market, consumer QoE becomes the main differentiator.

ELECTRIC POWER GRIDS ARE ONE OF THE VERTICALS THAT WILL BENEFIT MOST FROM M-QoE

At the same time, electric power grids and distribution systems are also under strain due to the growing introduction of non-centralized renewable energy resources, electric storage systems and increased electrical demand from new sources such as electric vehicles. The evolving needs of power grids are driving distribution systems to be more dynamic, with support for bidirectional power flows compared with the traditional centrally managed systems [4].

All of these factors introduce new challenges in the operation, planning, protection and control of future power grids. The growing need for new solutions is being met by the introduction of smart grids enabled by advanced communications networks including smart devices, wireless connectivity, automation and real-time control, edge computing and analytics.

To improve power quality, many electrical grids are being upgraded with synchrophasor technology. With this in mind, we decided to test our proposed M-QoE framework by applying it in a smart-grid scenario to predict the M-QoE of devices called phasor measurement units (PMUs) in a cellular-based synchrophasor.

Synchrophasors

A synchrophasor is a time-synchronized measurement of a quantity described by a phasor [5, 6]. Like a vector, a phasor has magnitude and phase information. PMUs in a traditional PMU network are deployed across transmission grids to measure voltage and current, and with these measurements calculate parameters such as frequency and phase angle.

PMU measurements are time-stamped to an accuracy of a microsecond, synchronized using the timing signal available from GPS satellites

or other equivalent time sources. Measurements taken by PMUs in different locations are therefore accurately synchronized with each other and can be time-aligned, allowing the relative phase angles between different points in the system to be determined as directly-measured quantities. Synchrophasor measurements can thus be combined to provide a precise and comprehensive view of an entire interconnection.

A typical synchrophasor network is composed of PMUs and phasor data concentrators (PDCs) that are the control centers of the wide area measurement system (WAMS). The communication systems play a crucial role in the performance of the WAMS. Requirements for data reporting rates are typically 30 to 60 records per second, and may be higher; in contrast, current SCADA (supervisory control and data acquisition) systems often report data every four to six seconds – over 100 times slower than PMUs.

Multiple communications technologies are used in synchrophasors, from power-line communications to microwave, but in order to meet the latency requirements, optical fiber is widely used. Traditional optical fiber solutions may become prohibitively expensive, though, as smart grids become more complex and the need for additional measurements in distribution grids grows. In light of this, work is underway to develop a new, micro-PMU (μPMU). A μPMU is a smaller, lower-cost and more accurate version of a PMU that can be deployed more extensively in a distribution grid and provide accurate information from more places, allowing for better dynamic distribution grid management.

Cellular-based synchrophasors

The cost of extending the WAMS to connect more devices can become an issue when deploying μPMUs at scale in distributed grids. 5G with network slicing can meet the connectivity requirements of a WAMS and reduce the cost of deploying μPMUs. Using low-band and mid-band radio frequencies, the data bandwidth requirements of PMU applications can be achieved, with latency values in access networks (radio access to network edge) of 10ms.

A μPMU IS A SMALLER, LOWER-COST AND MORE ACCURATE VERSION OF A PMU

IoT Connectivity as a Service (CaaS) is a subscription-based solution offered by operators that utilities can purchase to manage the connectivity of their PMUs and μPMUs.

Private LTE is another cost-effective alternative for grid operators to build synchrophasors over a dedicated communication system managed by the mobile operator to meet the desired requirements of latency, reliability and coverage.

Synchrophasors – Machine QoE assessment

According to a recent report from the North American SynchroPhasor Initiative (NASPI) [4], PMU applications in a power system can be classified into four main categories based on their reliability ranges:

- » automation (latency of ≤ 10-100ms)
- » reliability (latency of ≤ 1,000ms)
- » planning (latency of ≤ 100-1,000ms)
- » operation (latency of ≤ 1,000ms).

The report also lists the relevant KPIs – accuracy, reliability, latency, and message rate – and ranks them per category on an importance-level scale of 1-4 in which 4 means critical, 3 means important, 2 means somewhat important and 1 means less important.

Table 2 presents the key details of the NASPI’s classification of PMU applications in power systems.

Automation refers to the automated protection and control applications of PMUs in distribution systems. The aim of these applications is to improve the reliability and security, automated remedial action schemes, and asset utilization. Reliability is a class of PMU applications that can be divided into topology and disturbance detection and situational awareness. Planning is a class of PMU applications focused on better system understanding and improved system modeling of the distribution network.

Application	Accuracy	Availability/reliability	Low latency	Message rate
Automation	4	4	4	4
Reliability	2	2	3	2
Planning	4	3	1	4
Operation	1	1	2	2

Table 2 Classification of PMU applications in power systems

Operations is associated with the monitoring and visualization of the network performance.

A CSP supplying network connectivity services to a smart-grid operator can use the proposed framework for the assessment of customer M-QoE by calculating $M-QoE = \sum [\alpha_i * M-QoE_i]$ where:

- » i represents the relevant features of synchrophasor applications (accuracy, reliability, latency, message rate and security).
- » α_i represents the importance level coefficients of the KPIs derived from the number in Table 2 between 1 and 4.

We tested this M-QoE framework in an experimental 5G lab environment, where we built a synchrophasor network and connected between one and ten μPMUs to a PDC, depending on the test scenario. Probes were designed and installed in the communication network, in the μPMUs and in the PDC to collect and stream network and device O-KPI_i measurements associated with the transmission of payload data between μPMUs and the PDC.

To obtain enough O-KPI measurements, we identified the locations of potential μPMU devices in the grid network and used the M-QoE prediction model to generate the measurement data. This data was aggregated in the knowledge-based system running the rules-based and ML models for continuous evaluation and monitoring of the overall M-QoE of the grid operator. The experiment confirmed that the framework will assist CSPs

in monitoring and predicting the M-QoE of an enterprise with widespread devices over their infrastructure without the need for expensive large-scale measurements.

In another experiment, we used the M-QoE framework for the measurement of packets lost and delay gaps in the transmission of synchrophasor data due to communication link failures. The results showed early prediction of failures and M-QoE as well as accurate detection of failure types [7].

Conclusion

The deployment of 5G networks will accelerate the growth of the Internet of Things (IoT) and offer a wealth of new business opportunities for communications service providers (CSPs). It is important to recognize, however, that the churn of a single IoT enterprise due to network issues may represent many device connections with significant revenue impact. To ensure consistent QoE in IoT applications, CSPs need a smart solution to consistently monitor the diversity of performance requirements that are requested by IoT subscribers. Ericsson’s Machine QoE framework is designed with this in mind, utilizing the power of artificial intelligence and machine learning to automatically discover and predict events in the CSP network, so that they can be addressed before they have a negative impact on QoE.

References

1. Qualcomm, Making 5G NR a reality, December 2016, available at: <https://www.qualcomm.com/media/documents/files/whitepaper-making-5g-nr-a-reality.pdf>

2. Elsevier, Pervasive and Mobile Computing, vol. 48, pp. 59–68, 2018, Traffic characterization and LTE performance analysis for M2M communications in smart cities, Malandra, F; Chiquette, L.O; Lafontaine-Bédard, L.-P; Sansò, B, available at: <https://www.sciencedirect.com/science/article/abs/pii/S1574119217306089>

3. Ericsson Technology Review, Generating actionable insights from customer experience awareness, September 30, 2016, Niemöller, J; Washington, N; Sarmonikas, G, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/generating-actionable-insights-from-customer-experience-awareness>

4. Energies 2019 vol 12, A Comprehensive Survey on Phasor Measurement Unit Applications in Distribution Systems, November 29, 2019, Hojabri, M; Dersch, U; Papaemmanouil, A; Bosshart, P, available at: <https://doi.org/10.3390/en12234552>

5. North American SynchroPhasor Initiative (NAPSI), Synchrophasor Technology Fact Sheet, October 2014, available at: https://www.naspi.org/sites/default/files/reference_documents/33.pdf?fileID=1326

6. IEEE, Micro-synchrophasors for distribution systems, ISGT 2014, May 19, 2014, von Meier, A; Culler, D; McEachern, A; Arghandeh, R, available at: <https://ieeexplore.ieee.org/document/6816509>

7. IEEE, A New Approach to Reliability Assessment and Improvement of Synchrophasor Communications in Smart Grids, May 12, 2020, Seyedi Y; Karimi H; Wetté C; Sansò B, available at: <https://ieeexplore.ieee.org/document/9091616>

Further reading

» Ericsson, Internet of Things, available at: <https://www.ericsson.com/en/internet-of-things>

» Ericsson, 5G, available at: <https://www.ericsson.com/en/5g>

» Ericsson, Network slicing, available at: <https://www.ericsson.com/en/digital-services/trending/network-slicing>

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Versatile Video Coding explained

– THE FUTURE OF VIDEO IN A 5G WORLD

Many new and emerging 5G use cases will soon require video compression efficiency and functionality that are beyond the capabilities of today's leading video codecs. Versatile Video Coding – the new video compression coding standard recently approved by the Moving Picture Experts Group and the International Telecommunication Union – includes both improved compression efficiency and new features to enhance support for immersive video and low-delay video coding.

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The latest Ericsson Mobility Report estimates that video accounted for 63 percent of the traffic in mobile networks during 2019 and that this share will increase to 76 percent by 2025 [1]. Enabled by continuously improving network performance, this growth is driven by the ever-increasing availability of video content in streaming services and online apps, and by changing consumer behavior toward consumption of the content on mobile devices. The evolution toward large screens with high resolutions beyond HD on mobile devices further raises consumer expectations regarding the quality of delivered video content. Video codecs play a critical role in coping with these expectations.

While the quality of video content depends on several characteristics such as high pixel bit depth, high frame rate, wide color gamut (WCG) and high dynamic range (HDR), it is the resolution (the number of pixels in a video picture) that is most directly associated with the bandwidth required for transmission. Other key factors determining the required bandwidth are related to the type of the video content and the latency with which the content is delivered to the end user.

At the same time, innovations in 5G networks offer new opportunities for video-enabled services for both consumers (remotely rendered virtual/extended reality and cloud gaming, for example) and industries, particularly with respect to the Internet of Things (IoT) and the automotive sector.

These new services are expected to rely on continued video evolution toward 8K resolutions and beyond, and on new strict requirements such as low end-to-end latency for video delivery. Since the data rate of such video-enabled services is extremely high, the cost would be prohibitive – even in the most modern networks – unless the video was carried in a compressed format, with the help of a next generation video codec.

Versatile Video Coding (VVC), the new video compression coding standard that will be published as ISO/IEC 23090-3 and ITU-T recommendation H.266, offers the highest compression efficiency available today and is therefore the codec that is best suited to offer a suitable performance level for new media services over 5G networks. It can also enhance the user experience for existing video services by delivering substantially higher quality at the same bitrate.

Alternatively, it can be used to reduce the bitrate (roughly by half) while maintaining the same quality. For some legacy applications, the cost of replacing an older codec such as H.264 with a newer one such as

HEVC has not been economically beneficial, since the reduction in bit rate of 40 percent has not compensated for the cost of replacing a large number of set top boxes. Even in some of these cases, the introduction of VVC may change this calculation since the bit rate reduction compared to the older codec is so much greater. *Figure 1* provides an overview of potential application areas.

Video compression options available today

A video codec, which can be implemented in hardware or software, encodes and/or decodes digital video. Modern video codecs can reduce the bitrate of uncompressed video to less than one percent of the original rate without any noticeable visual quality degradations. The majority of these video codecs have been standardized and published by a standards development organization (SDO).

The most popular and widely deployed codecs include the MPEG (Moving Picture Experts Group) series from ISO/IEC (International Organization for Standardization/International Electrotechnical Commission) and the corresponding H.26x series

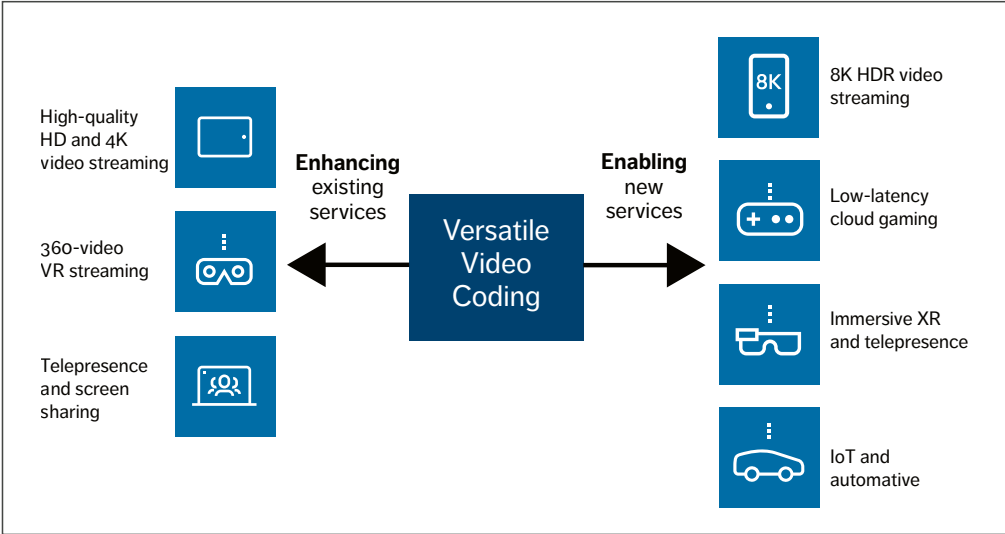


Figure 1 VVC application areas

from the ITU-T (ITU Telecommunication Standardization Sector), which were developed jointly. The naming convention for these codecs is MPEG-2/H.262, AVC/H.264 (Advanced Video Coding) and HEVC/H.265 (High Efficiency Video Coding). Other video codec options include Google's VP8 and VP9, and the Alliance for Open Media's AV1.

●● VVC IS THE LATEST MEMBER OF A SUCCESSFUL FAMILY OF VIDEO CODECS ●●

MPEG-2/H.262

The MPEG-2 video codec was published in 1994 and is still in wide use today in standard definition digital TV services. Its compression efficiency is low when compared with more recent video codecs. The main reason for using MPEG-2 today is to support existing set-top-boxes, as the cost of replacing these may be higher than any potential savings from a new video codec.

Advanced Video Coding

AVC/H.264 was published in 2003 and is currently the most widely used video codec. It is supported by practically all mobile devices, is heavily used for video carried over the internet, and is the preferred video codec for HDTV.

High Efficiency Video Coding

HEVC/H.265 [2] is the successor to H.264 and was published in 2013. Compared with H.264, it delivers the same visual quality at roughly 40 percent lower bitrate. There is widespread support for HEVC across TVs and mobile devices. HEVC has been selected by the DVB (Digital Video Broadcasting) and ATSC (Advanced Television Systems Committee) standards organizations for 4K broadcast services and is recommended by 3GPP for HD (HDR) and 4K mobile streaming.

VP8 and VP9

On2 Technologies, which was acquired by Google

in 2010, released the VP8 video codec in 2008, which then became a proposed royalty-free option in 2013. It has mostly been used as an alternative to H.264 in WebRTC, a framework for real-time web communication. VP9, a successor to VP8, was developed internally at Google and released as open source in late 2012. The use of VP9 for 4K YouTube videos has led to TV manufacturers incorporating VP9 decoding into virtually all 4K TV sets, thereby spreading support of VP9 to a substantial number of devices.

AV1

In 2015, a group of companies including Amazon, Facebook, Google, Intel, Microsoft and Netflix founded the Alliance for Open Media (AOMedia) with the aim of developing a new royalty-free video codec. The specification for that codec, AV1, was published on January 8, 2019. Although patent reviews were conducted during the development to avoid infringing on third-party intellectual property rights, the patent licensing organization Sisvel announced in March 2019 that it would form a patent pool for AV1. As a result, the royalty-free status of AV1 is uncertain.

Versatile Video Coding – the development process

Versatile Video Coding (H.266) was standardized in a joint effort by the Video Coding Experts Group of the ITU-T and MPEG of the ISO/IEC and is therefore the latest member of a successful family of video codecs that includes MPEG-2, H.264 and HEVC. In contrast to proprietary alternatives, all four of these video coding standards were developed in an open and collaborative fashion with agreed requirements and timelines.

The development of VVC followed a well-established standardization process, which started with a technology exploration activity in 2015, included a formal call for proposals in 2018 and concluded in July 2020 as a technically frozen standard. The development process involved international stakeholders across the entire media ecosystem: content producers, manufacturers,

operators, broadcasters, chipset vendors and academics. Throughout the process, all documentation including technical contributions, draft specification text, reference software and conformance bitstreams were made available publicly.

Ericsson has been an active participant in video standardization for more than 20 years and was closely involved in the development of the VVC standard. Throughout the process, we led several of the core experiments, chaired ad-hoc working groups and made significant contributions to the development of the technology behind the video codec, most notably in the areas of deblocking filtering, reference picture management, low-delay video coding and optimized encoder configurations. We also participated in efforts that made an impact in other areas of the video codec such as intra and inter prediction and adaptive loop filtering.

Key benefits of versatile video coding

As the latest and most sophisticated video codec to date, VVC offers the highest compression efficiency of all video codecs, and it is particularly appropriate

for higher resolution video streams due to its coding tools, which can operate on block sizes of up to 128x128 pixels and with 64x64 sample size transforms. VVC can achieve a reduction in bitrate of around 40 percent for existing HD and 4K services deployed with HEVC, at the same visual quality.

Figure 2 provides a performance comparison between VVC and four other video codecs. The comparison was done by Ericsson Research using various sources of information including in-house testing [3, 4]. The figure shows the approximate relative bitrate requirement for each video codec in order to reach the same video picture quality for HD and 4K content, benchmarked against the HEVC video codec, which is normalized to 100 percent. VVC performs significantly better, requiring only 60 percent of the relative bitrate compared with both HEVC and AV1.

On top of its superior compression performance, the versatility of VVC also makes it an attractive choice beyond mainstream 2D video services. VVC has been designed to handle both traditional camera-captured content as well as the increasingly

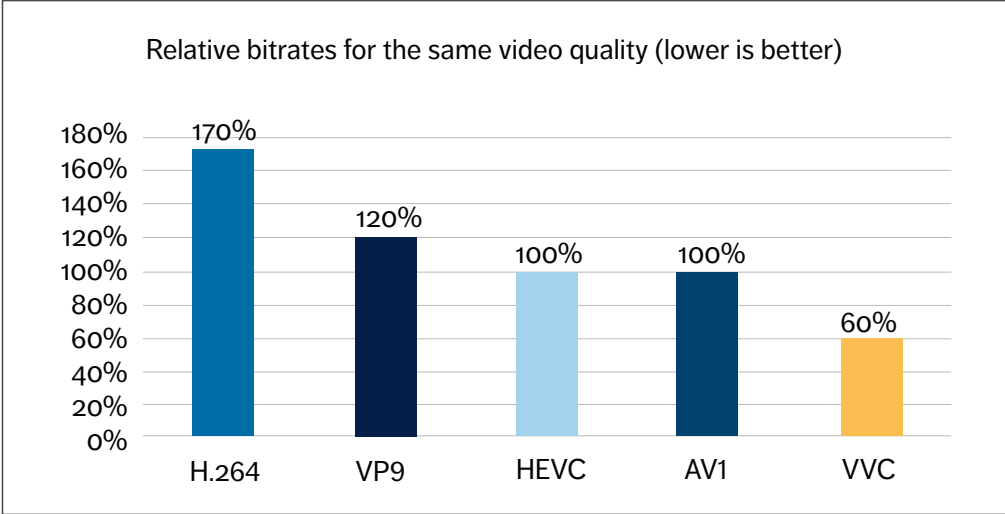


Figure 2 Relative bitrates for the same video quality

prevalent computer-generated imagery used in applications such as online gaming, e-sports video streaming and screen-sharing applications.

Immersive video

In contrast to traditional two-dimensional video, where one particular view is captured by a camera, immersive video is video in which every angle is recorded. During playback, which at present typically occurs on a virtual reality (VR) headset, the user is not constrained to a particular view but can look around freely. When the viewer does not change position, this is called three degrees of freedom (3DoF) immersive video, or more commonly 360-degree video.

360-degree video is typically stored in a projection format. One such format is equirectangular projection, which is similar to how the earth is pictured on a world map. A more popular format, however, is cube map projection. Here, each face of the cube represents one-sixth of the sphere area, and the six faces are arranged in one rectangular video picture. This enables the use of two-dimensional video codecs to handle the immersive content.

Due to the nature of the human visual system, a high-quality representation of the full 360-degree video sphere is only needed for the user’s current gaze direction. A well-known technique for exploiting this property is to split the full 360-degree video content into multiple small rectangular regions and only convey and render regions in high quality that cover the area the user is currently looking at, called the viewport. To accommodate fast head motion, video is also transmitted for the non-viewport area, but the quality of that video is much lower.

To that end, and to help with other immersive video applications such as remotely rendered VR, VVC introduces the concept of subpictures, which allows for the efficient extraction and merging of pieces of video of different quality that is a requirement for larger resolution immersive formats. A subpicture is a rectangle within the full video picture that is fully independent of other subpictures

and designed to be easy to extract and merge with other subpictures. Previous video codecs provided similar functionality using motion-constrained tile sets, but subpictures are both much more efficient and easier to manage.

The efficiency of the subpicture design comes from built-in adjustments of low-level coding tools, where previous designs require the encoders to be very restrictive. Previous designs also require rewriting of substantial amounts of coded data – a burden that is largely alleviated by the use of subpictures in VVC. The use of subpictures in VVC significantly reduces the complexity of application systems. This capability, together with its superior compression efficiency, makes VVC the best video codec choice for immersive video applications.

THE USE OF SUBPICTURES IN VVC SIGNIFICANTLY REDUCES THE COMPLEXITY OF APPLICATION SYSTEMS

Low-delay video coding

Low-delay video coding is a key technology for certain time-critical video applications such as videoconferencing, cloud gaming and remote control of road vehicles and drones. To reduce the latency caused by excessive data buffering, the video encoder typically aims to generate as smooth a data rate as possible. One well-known video codec feature is gradual decoding refresh (GDR), which provides smooth tune-in points in the bitstream without the need to encode single refresh pictures (intra pictures) that result in bitrate spikes and high latencies. Instead, smooth tune-in points are generated by spreading the refresh across multiple pictures.

GDR has been supported in many older video coding standards such as H.264 and HEVC, but in those cases the feature is conveyed in a supplemental enhancement information (SEI) message, which makes GDR optional. This means that an encoder cannot be certain that the decoders are capable

of tuning in at GDR positions. However, during standardization of VVC, an Ericsson proposal to add a GDR picture-type indicator to the standard was adopted. This makes GDR support mandatory in VVC, which ensures decoder interoperability for such low-delay applications.

Besides the GDR feature, VVC also inherited decoding unit (DU) based decoder operation from its predecessor, HEVC. DUs enable an encoder to output a first part of a picture, in a specified controlled manner, without requiring the entire picture to first be encoded. DUs and mandatory GDR support make VVC the prime video codec choice for low-delay video applications.

How VVC works

It is important to note that the superior coding efficiency of VVC is not due to any single compression tool. Rather, it is the result of combining many tools, each contributing with a small compression improvement. The most significant of these tools are block partitioning, advanced inter prediction, dependent quantization, adaptive loop filtering and improved deblocking filtering. The addition of sophisticated new coding tools has also had an impact on computational complexity. As a result of these developments, it is expected that VVC decoder complexity will be around twice that of HEVC.

Block partitioning

Similar to its predecessors, VVC uses a block-based hybrid coding architecture. A video codec based on

such an architecture does not code pixel values directly but instead predicts and codes only error information to compensate for the inaccurate prediction. The process runs on blocks of pixels rather than the entire picture, since this makes it possible to adjust to local video picture characteristics, minimizing the prediction error. For video codecs based on such an architecture, an efficient picture partitioning scheme is pivotal to achieving high compression efficiency.

In VVC, each picture is split into non-overlapping squares called coding-tree units (CTUs). The largest CTU size allowed in VVC is 128x128 pixels, larger than the 64x64 maximum size allowed in HEVC. Large blocks improve the efficiency of coding flat areas such as backgrounds, especially for high-resolution videos such as HD and 4K. In order to efficiently represent highly detailed areas such as textures and edges, VVC employs a flexible partitioning scheme that can partition 128x128-sized CTUs down to coding units (CUs) as small as 4x4 pixels.

The scheme is based on two parts. The first is the quaternary tree (quad tree) split that is also available in HEVC, which can recursively split CTU into squared CUs down to 4x4 pixels, smaller than the 8x8 minimum CU size in HEVC. The second part consists of binary-tree and ternary-tree splits that partition a block into two and three rectangles respectively. Both binary and ternary tree splits can operate in either horizontal or vertical directions, be recursively applied and mixed together in a nested multi-type tree.

Terms and abbreviations

3DOF – Three Degrees of Freedom | ALF – Adaptive Loop Filter | CTU – Coding-Tree Unit | CU – Coding Unit | DU – Decoding Unit | DVB – Digital Video Broadcasting | GDR – Gradual Decoding Refresh | HDR – High Dynamic Range | HEVC – High Efficiency Video Coding | IEC – International Electrotechnical Commission | IETF – Internet Engineering Task Force | ISO – International Organization for Standardization | ITU – International Telecommunication Union | ITU-T – ITU Telecommunication Standardization Sector | MPEG – Moving Picture Experts Group | SDO – Standards Development Organization | UHD – Ultra High Definition | VR – Virtual Reality | VVC – Versatile Video Coding | XR – Extended Reality

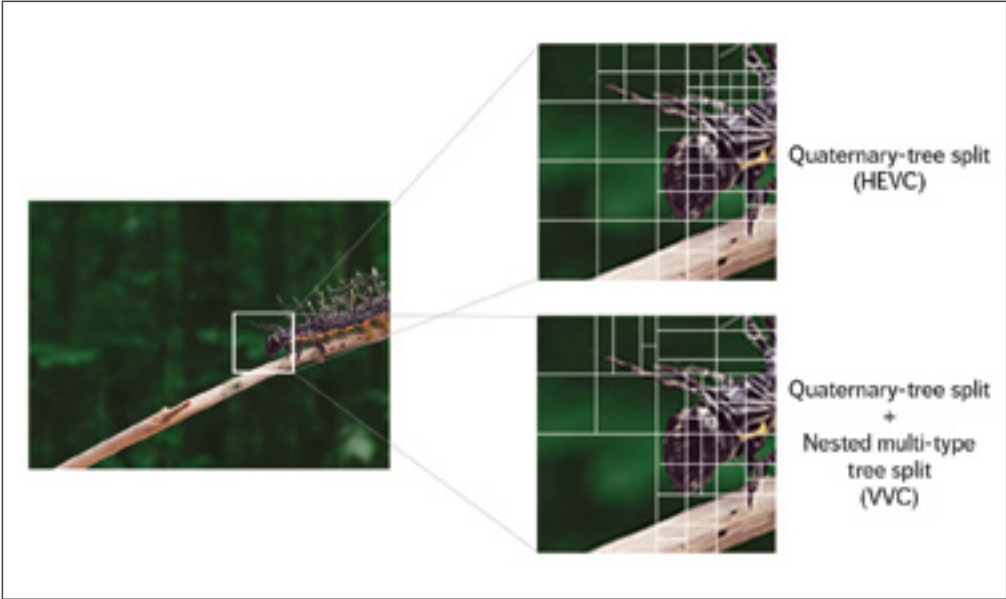


Figure 3 Comparison of the partitioning of a close-up section of a video picture in HEVC and VVC

Figure 3 provides an example of the partitioning of a close-up section of a picture. HEVC block partitioning (at top right) uses quaternary-tree split with coding blocks up to 64x64 pixels. VVC block partitioning (at bottom right) uses quaternary-tree split with coding blocks up to 128x128 pixels and nested multi-type tree split employing binary and ternary tree splits.

The block partitioning in VVC is highly flexible and provides about 8 percent bitrate reduction over HEVC [5]. However, this flexibility comes at a computational cost, especially on the encoder side, where many more permutations need to be evaluated to select the optimal partition.

Advanced inter prediction

One of the most efficient bit-saving techniques in video compression is inter-picture prediction (more commonly referred to as inter prediction), which simply means copying sample values from previously coded pictures. The information about where to

copy from – the horizontal and vertical displacement for a block – is stored in what is called a motion vector. Techniques that improve inter prediction are responsible for a substantial part of the bitrate reduction between VVC and HEVC.

While many techniques have been used to achieve the advanced inter prediction in VVC, five of them are particularly noteworthy: the affine motion model, adaptive motion vector resolution, bi-directional optical flow, decoder side-motion vector refinement and geometric partitioning mode. The precision of the motion vectors has also been increased to 1/16 pixel compared to the quarter pixel resolution of HEVC.

In previous video coding standards, it has been possible to compensate for translational motion; that is, the decoder can be instructed to fetch sample values not from the same place in a previous image but rather from another position (1.75 pixels to the left, for example). In VVC, the affine motion model tool makes it possible to specify not only distance

but also rotation and zoom, which can save a lot of bits for sequences containing such motion.

Another bit-saving technique in VVC is the ability to vary the precision of the motion vectors. For example, the encoder can signal to the decoder that the incoming motion vectors are in integer or four-times-integer resolution. This can save bits when predicting smooth areas where the exact sub-pixel precision does not give much image improvement over integer precision and also when representing large motion vectors.

The advanced inter prediction in VVC further exploits the fact that in some cases motion information does not have to be explicitly transmitted (as motion vectors) but can instead be inferred from similarly moving parts of the picture. For example, the bi-directional optical flow tool in VVC can infer motion vectors by measuring the optical flow in reference pictures and the decoder side-motion vector refinement can infer motion vectors by minimizing differences between reference pictures.

Finally, the advanced inter prediction in VVC also exploits motions of non-rectangular shapes to better align with the shape of moving objects. This means that one half of the block can have one motion vector and the other half of the block can have another with the halves separated by a single geometrical line that is determined by an angle and an offset.

Dependent quantization

The quantizer is a core part of a video codec that is directly linked to operational control of video bitrate and visual quality. By adjusting the quantization step, the encoder controls the fidelity of the error signal (transform coefficients), which is then coded with an entropy coder and sent in the video bitstream.

Previous standards such as HEVC have used a scalar quantizer but since the entropy coder processes each transform coefficient in a coding block in sequential order, there was an inherent inefficiency if the quantization level of each coefficient were to be determined independently.

To optimize the determination of quantization levels within a block, sophisticated encoders use an algorithm called trellis quantization.

In VVC, dependent quantization has been introduced, where the codec can switch between two shifted quantizers and thereby reduce the quantization error. In the VVC design, the quantization levels for a given transform coefficient depend on the values of the preceding quantized coefficients. To use this tool effectively, an encoder therefore needs to evaluate how the determined quantization level for each coefficient impacts both the bit count and the total reconstruction error for the whole block. In this way, the dependent quantization removes the inefficiency of quantizing coefficients independently and thereby provides a substantial bit-reduction over HEVC.

DEPENDENT QUANTIZATION REMOVES THE INEFFICIENCY OF QUANTIZING COEFFICIENTS INDEPENDENTLY

Adaptive loop filtering

The adaptive loop filter (ALF) is a new in-loop filter in VVC. ALF scans the picture after it has been decoded and selectively applies (on a CTU basis) one of several two-dimensional finite impulse response filters before the picture becomes output or is used for prediction. The video encoder calculates the sets of filter coefficients that will lead to the smallest error and transmits those to the video decoder. ALF has the ability to clean up artifacts in the picture and also contributes to a substantial bit-reduction over HEVC.

Improved deblocking filtering

VVC and its predecessors HEVC and H.264 are all block-based video codecs. The downside of the block-based approach is that it can give rise to “block artifacts” – visible edges at some block borders. Deblocking filtering is an approach to reduce these artifacts by selectively smoothing

across the block boundaries. The deblocking filtering in VVC is based on the HEVC deblocking filtering, for which Ericsson was the leading contributor. On top of this already strong base, VVC is capable of using longer deblocking filters, where major parts were designed by Ericsson. The long deblocking filters allow for stronger deblocking that can be more effective in hiding block artifacts, especially for larger blocks (128x128 sized blocks, for example) in relatively smooth areas. The long deblocking filters contribute significantly to the improved subjective quality of VVC compared to HEVC.

THE LONG DEBLOCKING FILTERS CONTRIBUTE SIGNIFICANTLY TO THE IMPROVED SUBJECTIVE QUALITY OF VVC COMPARED TO HEVC

What's next for VVC?

In today's competitive video codec landscape, video compression performance is key to successful market adoption, but it is not the sole determining factor. Availability of the technology is critical, especially the availability of hardware-accelerated decoders in chipsets and processors. A recent prediction from a major chipset vendor stated that first commercial VVC shipments could start as soon as 2021. Software-based solutions are typically faster to roll out and are essential especially in the early phases of deployment across the ecosystem. In the case of VVC, the first demonstrations of real-time software decoders took place shortly before the standard was completed.

Since video codecs do not operate in a silo, support and interoperability across the ecosystem in terms of media delivery protocols and application specifications are required. Some of the SDOs such as MPEG and IETF are looking into providing support for carriage of VVC in their respective

media transport specifications. Organizations such as 3GPP and DVB are investigating VVC in the context of next-generation services including 5G-enabled ones such as 8K (7680x4320 video) and 360-degree video VR streaming. Unlike its predecessors, the first version of the VVC standard includes support for a broad range of applications across the media ecosystem, which is likely to have a positive impact on the cost of deployment and interoperability of VVC-based solutions and services.

In order to facilitate cross-industry discussion around non-technical aspects of VVC deployment such as licensing, marketing and interoperability activities, Ericsson and other industry leaders launched the Media Coding Industry Forum (MC-IF) in 2018. Since then, MC-IF has hosted a series of workshops and events to gather wider industry input on commercial aspects that may further early VVC adoption. In particular, timely availability of licensing terms for VVC was established as one of the key factors for VVC deployment. To this end, shortly after the finalization of VVC development, MC-IF initiated fostering efforts for a patent pool program essential to VVC.

Conclusion

Versatile Video Coding represents state-of-the-art video coding and is certain to play an important role in supporting a wide range of 5G use cases. Designed to manage the high demand that increasing amounts of video poses on networks, our research reveals that VVC achieves the best available compression performance at a computational complexity suitable for implementation in both software and hardware. As one of the main contributors to VVC, we believe its deployment will significantly reduce the data rates of existing video services as well as serving as a primary enabler for next-generation media services.

References

1. Ericsson Mobility Report, June 2020, available at: <https://www.ericsson.com/en/mobility-report/reports/june-2020>

2. Ericsson Technology Review, Next generation video compression, April 24, 2013, Fröjd, P; Norkin, A; Sjöberg, R, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/next-generation-video-compression>

3. 2019 Picture Coding Symposium (PCS), Ningbo, China, 2019, pp. 1-5, Compression Performance of the Versatile Video Coding: HD and UHD Visual Quality Monitoring, Sidaty, N. et al., available at: <https://ieeexplore.ieee.org/document/8954562/authors#authors>

4. ResearchGate, SMPTE Motion Imaging Journal, vol. 128, no. 10, pp. 14-24, Nov.-Dec. 2019, Analysis of Emerging Video Codecs: Coding Tools, Compression Efficiency and Complexity, J. L. Tanou and M. Blestel, available at: https://www.researchgate.net/publication/332073033_Analysis_of_Emerging_Video_Codecs_Coding_Tools_Compression_Efficiency_and_Complexity

5. IEEE Transactions on Circuits and Systems for Video Technology, vol. 30, issue 5, pp. 1311-1325, May 2020, doi: 10.1109/TCSVT.2019.2945048, A VVC Proposal With Quaternary Tree Plus Binary-Ternary Tree Coding Block Structure and Advanced Coding Techniques, Y. Huang et al., available at: <https://ieeexplore.ieee.org/document/8859290>

Further reading

» Ericsson blog, Cutting the bitrate with Versatile Video Coding, available at: <https://www.ericsson.com/en/blog/2019/7/mpeg-cut-the-bitrate-versatile-video-coding>

» Ericsson blog, Industry Forum to promote Versatile Video Coding, available at: <https://www.ericsson.com/en/blog/2018/9/industry-forum-to-promote-versatile-video-coding>

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Cognitive processes for adaptive intent-based networking

Autonomously operated and self-adapting networks will make it possible to utilize the capabilities of 5G networks in new business models and achieve an unprecedented level of efficiency in service delivery. Intents will play a critical role in achieving this zero-touch vision, serving as the mechanism that formally specifies what the autonomous system is expected to do.

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5G networks introduce unprecedented flexibility and dynamic adaptation into service delivery and network resource utilization. In the business layer, this is reflected in the ability to offer customizable service products with detailed agreements on functional and non-functional characteristics as well as fast delivery. Dynamic adaptation to changes within the constraints of stringent requirements on lead and reaction times is beyond the capacity of a human workforce. Extensive automation will be necessary to overcome this challenge.

■ The zero-touch paradigm implies that the operation of services and the underlying networks is autonomous and does not require human intervention. To achieve this, the zero-touch system must be able to handle the complexity caused by continuous changes to the system at the same time that it delivers services to users and manages issues such as the cost of resources versus the budget available, the legal compliance of the service and the security of the setup. This is challenging for technical systems in real-world scenarios.

For example, successful service operation requires each service to be properly provisioned and assured to deliver the promised function with agreed

performance metrics. Complexity arises as a result of changes to contracts, products, customer preferences, the business strategy or the environment in which the service is being offered. Users may start exhibiting new behavior, leading to varying service usage patterns and network loads, or the network may change due to upgrades, reconfiguration or outages. Some changes may be regular and predictable, whereas others are sudden and surprising.

To manage all of these concerns autonomously and adapt its behavior appropriately, a zero-touch system must understand every aspect of what is expected of it. Each requirement and goal must be carefully defined in order for technical processes to derive suitable and optimized actions to manage it. These definitions are known as intents.

From the perspective of a human operator, an intent expresses the expectation of what the operational system is supposed to deliver and how it behaves. In light of this, we define intent as “formal specification of all expectations including requirements, goals and constraints given to a technical system.”

The role of intents in cognitive networks

Everything an autonomous system needs to know about its goals and expected behavior must be defined with intents. The system will not perform any operation unless it relates to the fulfillment and assurance of an intent, which means that all goals – including those that may have been considered “common sense” in human-operated systems – must be expressed as intents.

An intent in an autonomous system is ideally expressed declaratively – that is, as a utility-level goal that describes the properties of a satisfactory outcome rather than prescribing a specific solution. This gives the system the flexibility to explore various solution options and find the optimal one. It also allows the system to optimize by choosing its own goals that maximize utility.

Unlike traditional software systems, where requirements are analyzed offline to detect and resolve conflicts prior to implementation, intents are added to an autonomous system during runtime. Adaptation to changed intent as well as conflict detection and resolution are therefore essential capabilities of an autonomous system.

One of the benefits of expressing intents as utility-level goals is that it helps the system cope with the conflicting objectives of multiple intents. This is vital, because an autonomous system often has to take multiple intents into account before making a decision.

For example, an autonomous system may have one intent to deliver a service with high QoE, while another may be to minimize resource spending. It can resolve such conflicts either explicitly from weights that introduce relative importance or implicitly from properties of preferential outcomes as defined in utility-level goals.

Expectations originate from contracts or business strategy and remain constant when the underlying system is replaced or modified. Consequently, when setting up the intents, it is important that they are formulated in an infrastructure-agnostic way, so that they can be transferred across system generations and implementations.

Terms and abbreviations

AI – Artificial Intelligence | BSS – Business Support Systems | CEM – Customer Experience Management | IoT – Internet of Things | KPI – Key Performance Indicator | OSS – Operations Support Systems | RDF – Resource Description Framework | SLA – Service Level Agreement | SLO – Service Level Objective | SLS – Service Level Specification | SON – Self-Organizing Networks | TOSCA – Topology and Orchestration Specification for Cloud Applications | VNF – Virtual Network Function

In short, the intent establishes a universal mechanism for defining expectations for different layers of network operation. It expresses goals, utility, requirements and constraints. It defines expectations on service delivery as well as the behavior of the autonomous operational system and the underlying network.

☞ [THE INTENT] EXPRESSES GOALS, UTILITY, REQUIREMENTS AND CONSTRAINTS ☞

Service-specific intents

One essential type of intent relates to the specification of services. Service-specific intents state expected functional and performance characteristics. Service Level Agreements (SLAs), Service Level Specifications (SLs), Service Level Objectives (SLOs) and TOSCA (Topology and Orchestration Specification for Cloud Applications) [1] models are all examples of service-specific intents that are used on different levels in the operations stack.

SLAs are business support systems (BSS) objects. Service-specific intents based on SLAs specify the promised service and include expected performance details and business consequences such as payment for delivery and penalties when failing.

SLs/SLOs define the service delivery details at operations support systems (OSS) level. Based on this input, autonomous OSS would plan detailed tasks to realize the service delivery. TOSCA models would be used to express further technical details the OSS generate (expectations from orchestration and assurance).

In multiple stages, the autonomous operation makes decisions about further details. Higher-level intent is the input leading to the lower-level intent that is used to distribute specific goals to subsystems. For example, SLAs/SLs are the intents that express a terminal goal of the OSS. The OSS then decide which TOSCA model would be the best option to deliver the promised performance with minimal resource usage. The selected TOSCA

model is the instrumental goal of OSS and becomes an intent and terminal goal for the orchestrator.

This pattern of making decisions based on a given intent and taking action by sending lower-level intents to subsystems is the key interwork mechanism of intent-based operation, according to which the entire operations stack of autonomous networks is built.

Strategic and behavioral intents

Beyond all the service-specific intents that an autonomous system must have, it also requires guidance on how to handle strategic and behavioral concerns. Traditionally implemented in the form of manually coded policies, this type of guidance steers general system behavior and supports the type of decision-making that has traditionally been based on human intuition and experience, along with knowledge about context and operator strategy. Intent-based operation makes it possible for operators that want to handle these concerns in a more dynamic fashion to replace manually coded policies with strategic and behavioral intents.

This is useful in cases where the operator chooses to require a default minimum security level that differs from that which is implemented into the service, for example. In these cases, dedicated intent can be used to set the security level for all services that do not specify it directly.

With regard to legal compliance, services may be delivered in multiple markets where different rules apply. A legal-compliance intent requires compliance and potentially specifies the details.

Since there is always a risk of service degradation when changes are initiated and risky actions may sometimes lead to a higher margin, risk-management intents can be used to convey how the operator wants the autonomous system to balance risks versus potential gain.

Reporting/escalation intents steer how the autonomous system interacts with the human workforce by reporting progress status on intent fulfillment and seeking manual decision in escalations.

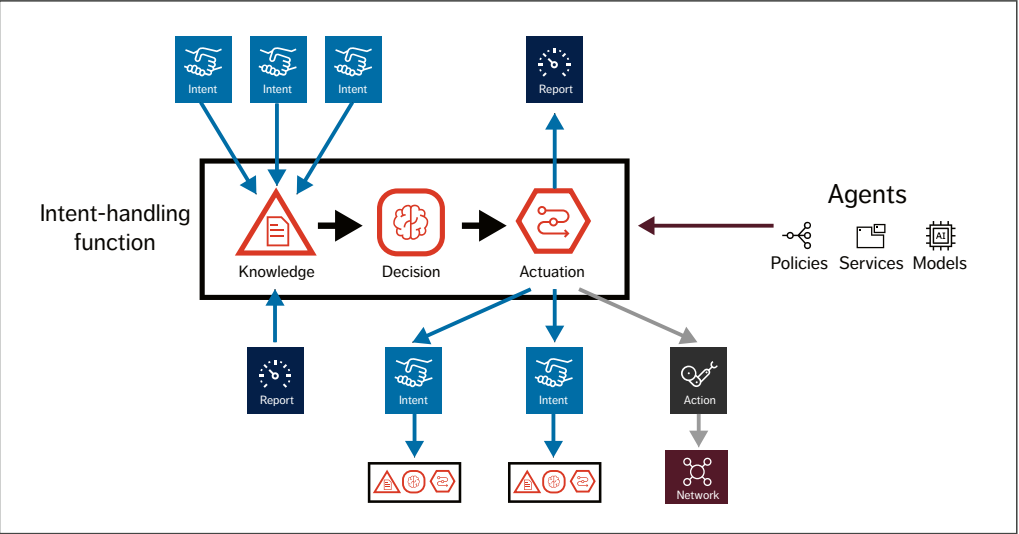


Figure 1 The intent-handling function

Formally expressing an intent

An autonomous system requires intents to be formally defined in a machine-readable and processable way, but the broad range of considerations involved and their abstract semantics are often difficult to structure. Techniques from knowledge management and semantic modeling enable the creation of an ontology of intent, based on an extensible metamodel. Resource Description Framework (RDF) [2] and RDF Schema [3] standards can be used for knowledge modeling.

Technical functions such as contract and order management would directly use RDF objects to communicate intent. Intent specified directly by human operators would require an intuitive frontend, potentially using natural language.

Intent handling

The operation of services within an intent-based network also requires the introduction of intent-handling functions in the operations stack and functional architecture. An intent-handling function receives the intents, decides which

actions must be taken to optimally fulfill all given intents and implements its decisions.

Intent-handling functions have a knowledge base that contains the intent ontology. They also have machine-reasoning capabilities to realize knowledge-driven decision-making processes.

Machine reasoning plays a key role in intent handling, with its capability to understand abstract concepts from diverse domains and provide precise, specialized conclusions based on precedent and observation. Probabilistic modeling contributes quantification of risk and uncertainty, which is essential to make informed decisions when facing conflicting goals and new situations.

Figure 1 shows how the intent-handling function works. While its implementation is domain-specific, its interface is generic. It receives intents that express all types of expectations. It is equipped with policies and artificial intelligence (AI) models that implement the capabilities needed for analyzing the system state and finding optimized operational actions based on observations from the operated environment. The intent handler also reports the fulfillment and assurance status of its intents.

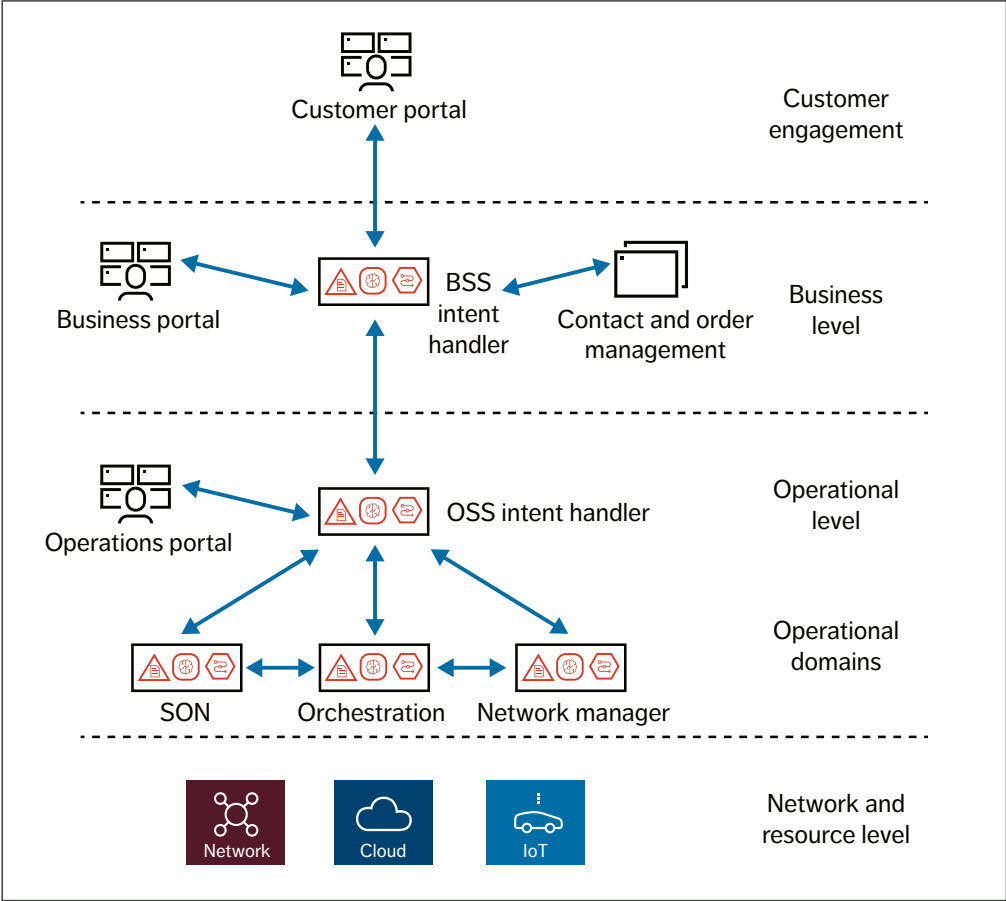


Figure 2 Intent-driven multi-layer operations system

The API (application programming interface) of the intent-handling function is domain-independent. Its main objective is to manage the life cycle of intents. It implements methods to set, modify and remove intents and send reports. Intent is constructed based on a common intent meta-model and its details are specified according to domain-specific information models. Intent management is therefore primarily knowledge management.

Figure 2 provides an example of how intent-handling functions can be combined to realize

a complete intent-driven operations system. Every major system layer and subsystem domain, including BSS, OSS, orchestration and network management, contains an intent-handling function. Intent originates from functions such as contract and order management. Additional intents can be entered directly through portals.

Introducing the cognitive layer

Lexico defines cognition as: "... the mental action or process of acquiring knowledge

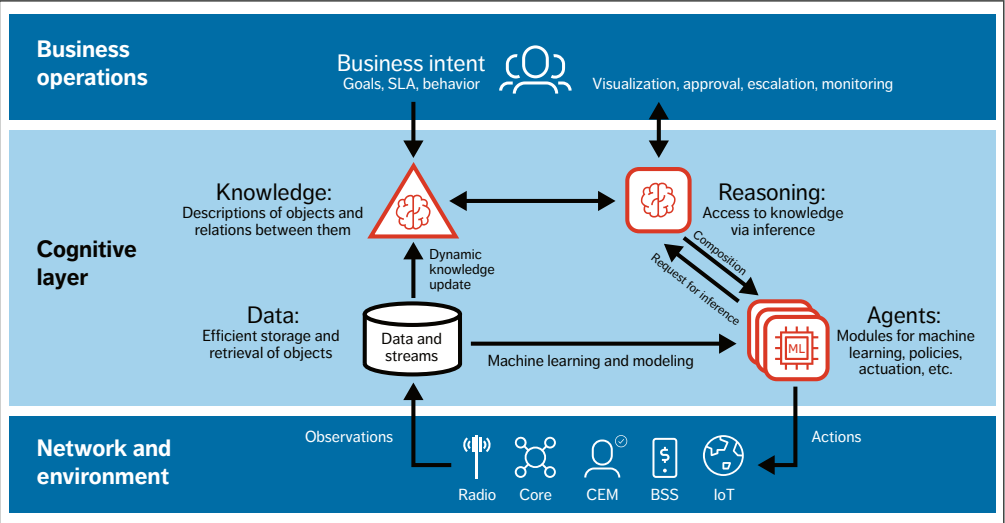


Figure 3 Functional architecture of the cognitive layer

and understanding through experience and the senses.” [4] As it is designed to perform the equivalent operational tasks of understanding through experiencing and sensing, an autonomous system is, therefore, a technical implementation of cognition.

Creating an intent-handling function that understands complex and abstract intent semantics, derives the optimal target state and plans actions for transitioning the system into this state is a challenging task. The function must be able to explore options, learn from precedents and assess the feasibility of actions based on their expected consequences.

By combining well-understood AI techniques within a flexible architecture, we have designed a cognitive system that specializes in autonomous service and network operation. We refer to it as the cognitive layer, and its role is to serve as an interface between business operations and the network/ environment, as shown in Figure 3.

The cognitive layer consists of three essential components: a knowledge base, a reasoning engine and an agent architecture. The knowledge base

contains the ontology of intents along with domain-specific knowledge such as the current state of the system. The domain-independent reasoning engine uses the knowledge graph and serves as the central coordinator function for finding actions, evaluating their impact and ordering their execution. Finally, the agent architecture allows any number of models and services to be used. Agents can contain machine-learned models or rule-based policies, or implement services needed in the cognitive reasoning process.

To be usable, an agent needs to be registered and described in the knowledge base. Its description can be added and modified at any time, allowing life cycles of the models, policies and supplementary services to be decoupled from the overall life cycle of the cognitive layer.

The agent metadata contains a description of the agent interface, along with its function, role and capabilities. For example, we have implemented a machine-learned model that can propose radio base station configurations that optimize the service experience. This model is registered as an agent in the role of a “proposer” for configuration actions.

The separate life cycle makes it possible for the model to be replaced with an improved version when available, independent of the cognitive layer release cycles.

We have also demonstrated agents in the role of “predictor” with the ability to estimate the effect of actions on key performance indicators (KPIs). An agent in an “observer” role would monitor data sources, keeping knowledge about the state up-to-date. An agent in the “actuator” role can implement actions in the network by utilizing, for example, established network management functions.

●● SPECIALIST AGENTS ARE USED INTENSIVELY IN EVERY STEP OF THE PROCESS ●●

How the cognitive layer works

The successful operation of the cognitive layer depends on smooth interaction between the reasoning engine and the knowledge base. The reasoning engine continuously executes a process that tries to find actions to close the gap between the current observed state and the wanted state, according to the intent. It collects proposals, obtains predictions on the effect of each proposal, evaluates gain versus risk and certainty, prioritizes actions and executes its decisions. Specialist agents are used intensively in every step of the process.

The reasoning engine is an adaptive knowledge-driven composer that can instantiate the cognitive process following changes in intent, state and context. It can dynamically compose specialized agents and add them into the process if their capabilities and roles match the needs according to intent and context. This is, for example, how the cognitive layer obtains action proposals from agents that are implementing suitable models.

In cases where the capabilities of multiple agents match the requirements of a role, each of them can be used simultaneously to generate alternative

solution strategies, resulting in a diverse set of options for the prediction and evaluation steps that follow. This coexistence of agents makes it possible to combine rule- and policy-based implementations with machine-learned alternatives in the same system, enabling the system to acquire new advanced abilities without losing current ones.

The cognitive process is a perpetual loop that starts again directly after the previous iteration has finished. Any degradation of the network or issues with services would be visible in the observed state. By trying to close the gap to the wanted state set by intent, the cognitive layer implicitly addresses incidents. Even without explicit issues, the continuous cognitive process would still seek actions for further optimization. It could, for example, try to deliver the same services with reduced resources.

Through its reasoning-based core process, the cognitive layer reaches a high degree of dynamic adaptability to new situations. This is in stark contrast to systems that have been realized through rule-based policies and fixed workflows, where every supported situation needs consideration at design time through suitable branches in the decision tree and diversifying rules. Existing rule-based policies can, however, still be used in the cognitive layer through integration as agents. This opens an upgrade path from legacy automation, with AI-based models added gradually.

Example use case

To demonstrate how the cognitive layer works, we have experimented with a use case that optimizes the provisioning decisions for the deployment of a virtual network function (VNF). The source of the intent is an SLA that specifies the service to be delivered along with the KPI targets that have been promised to the customer. Our example use case requires that the VNF be deployed with strict targets on latency and throughput.

We started by developing the required proposer and predictor agents in an offline data science process. We then deployed the service in a test environment and exposed it to a range of usage loads. We also explored deployment options by varying

parameters of the underlying TOSCA model. Combined with measured KPIs, these variations create training data sets for learning a model that is capable of connecting deployment options to expected performance. This exercise created a proposal agent that is able to recommend a TOSCA model configuration optimized for the latency and throughput figures required by the SLA.

In our experiment, online fulfillment – from receipt of the SLA to optimized deployment – is fully autonomous and controlled by the reasoning process of the cognitive layer. When the intent derived from the SLA arrives in the knowledge base, the reasoning loop starts processing it. The reasoner finds that the agent learned in the test environment matches the needs and requests a proposal. The agent delivers a TOSCA model fully configured and optimized for the requested KPI target and proposes to deploy accordingly.

The cognitive layer then uses prediction and evaluation agents to assess the proposal. In this case, our prediction agent contains a state-action model – a probabilistic graph based on Markov decision process modeling. The model is continuously learned from observing states and the results of observed actions. It has the ability to estimate the probability of expected result states for proposed actions, enabling informed decisions about actions considering their risk.

In the evaluation step, the autonomous system detects and resolves all remaining conflicts between intents. It also decides on escalations if the risk of the proposed deployment or uncertainty of the models is too high. If escalation is required, the cognitive process requests the support of a human technician, presents the situation including proposals and predictions, and asks for approval.

While the cognitive layer can operate fully autonomously, it knows when the human workforce wants to be involved. The exact threshold for escalation is at the discretion of the operator and is determined by behavioral intent. If the system gains human trust as a result of presenting many good actions, the threshold for fully autonomous decisions can be lowered.

When a proposed action is selected after verifying that the impact on all intents is advantageous and risk is reasonably low, it is then handed over for execution to the orchestrator behind the actuation agent. This concludes the first intent-handling loop after the new intent for the new SLA was introduced to the cognitive layer. The gap between the new intent and the current state was particularly wide since the new service was not yet provisioned. This gap narrowed through a provisioning action. Further iterations continuously monitor the service deployment, optimize it when possible or heal when needed. In this way, the continuous cognitive operation process provides fulfillment and assurance of expectations formulated as intents.

●● ONLINE FULFILLMENT IS FULLY AUTONOMOUS AND CONTROLLED BY THE REASONING PROCESS ●●

Conclusion

Zero-touch autonomy is an ideal beyond reach as long as artificial intelligence (AI) cannot match human capability to reason and decide within complex dependencies and broad domains. However, by using a combination of currently available and well-understood AI techniques within a flexible architecture, it is possible to reach a high degree of practical autonomous operation.

Our use case example demonstrates how the cognitive layer that we have developed can enable autonomous operation with the use of intents. It is built with an agent architecture that decouples the life cycles of the agents. It is coordinated by knowledge-driven reasoning that composes machine-learned models and legacy implementations. It evaluates action impact on intent fulfillment and makes decisions based on expected gains and risks, while resolving conflicting goals and maximizing utility. It can adapt to new situations through learning. The resulting system can execute many of the cognitive considerations human technicians

make when they operate services and networks manually. However, the cognitive layer does this with continuous attention and near-immediate reaction.

The adaptive reasoning capabilities of the cognitive layer enable effective management of growing network complexity, dramatically reducing

the need for humans to manually modify policies or participate in online decision-making. This frees up the human workforce to concentrate on offline tasks such as executing data science processes, optimizing the available models and defining business strategies that are implemented by intent setting.

References

- 1. **OASIS, TOSCA Version 2.0, Committee Specification Draft 02, June 25, 2020**, available at: <http://docs.oasis-open.org/tosca/TOSCA/v2.0/TOSCA-v2.0.html>
- 2. **W3C, RDF 1.1 Concepts and Abstract Syntax, W3C Recommendation, February 25, 2014**, available at: <https://www.w3.org/TR/2014/REC-rdf11-concepts-20140225/>
- 3. **W3C, RDF Schema 1.1, W3C Recommendation, February 25, 2014**, available at: <https://www.w3.org/TR/rdf-schema/>
- 4. **Cognition, Lexico.com**, available at: <https://www.lexico.com/definition/cognition>

Further reading

- » **Ericsson Technology Review, Cognitive Technologies in Network and Business Automation, 2018, Niemöller J; Mokrushin, L**, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/cognitive-technologies-in-network-and-business-automation>
- » **Ericsson, AI by Design**, available at: <https://www.ericsson.com/en/ai-and-automation>

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Artificial intelligence in RAN

A SOFTWARE FRAMEWORK FOR AI-DRIVEN RAN AUTOMATION

Artificial intelligence and its subfield machine learning offer well-established techniques for solving historically difficult multi-parameterization problems. Used correctly, these techniques have tremendous potential to overcome complex cross-domain automation challenges in radio networks. Our ongoing research reveals that an integrated framework of software enablers will be essential to success.

DIARMUID CORCORAN,
ANDREAS ERMEDAHL,
CATRIN GRANBOM

Modern telecommunications and mobile networks are becoming increasingly complex from a resource management perspective, with diverse combinations of software and infrastructure elements that need to be configured and tuned for efficient operation with high QoS.

■ The latest 5G mobile system is a good example of a sophisticated radio network that allows many deployment variations – such as centralized, distributed or various hybrids of both – while simultaneously supporting diverse categories of applications such as mission-critical control with ultra-reliability and low latency, massively

concurrent Internet of Things device access and enhanced mobile broadband. It is well accepted in the communications community that appropriately dimensioned, efficient and reliable configurations of systems like 5G are a complex technical challenge. Increased real-time, closed-loop, automation at all levels and time frames is a critical tool in controlling this complexity and ultimately reducing the capex and opex of radio network operations. There is currently a significant trend toward the use of artificial intelligence (AI) technology as an enabler for automating both repetitive and complex tasks. Applied correctly, AI and machine learning (ML) techniques have the potential to boost system

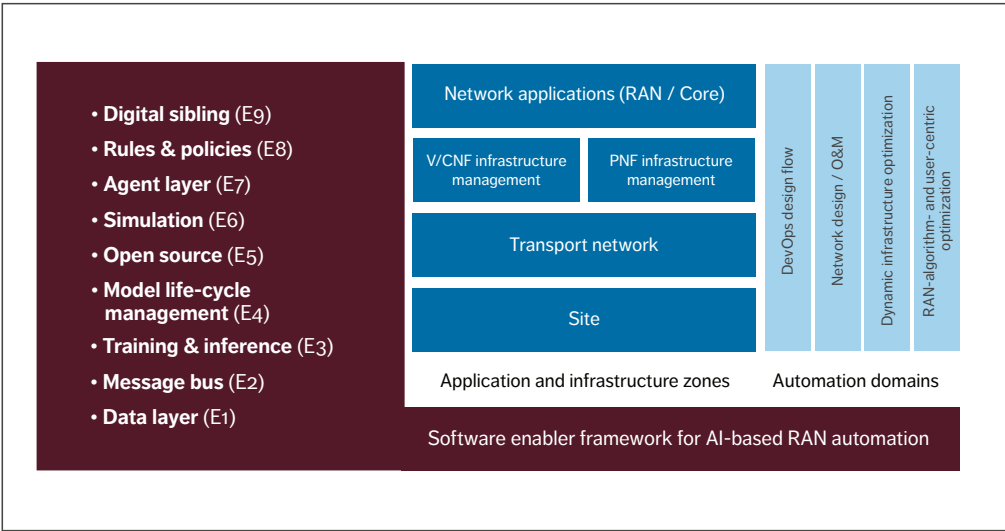


Figure 1 Software-enabler framework for AI-based RAN automation

efficiency by drastically simplifying certain types of automated management and control. It is important to realize, however, that this development will not occur without effort; on the contrary, the need for AI and ML techniques will have a profound effect on how the next generation of RAN should be designed and realized.

As part of this work, we have created a set of software enablers that target efficient and effective use of AI technologies in RANs. Together, they form an important framework of building blocks in realizing the next level of autonomous, AI-based, resource management in 5G and future-generation networks.

Landscape of automation in RAN

The wide variety of architectural and transport options that are possible in 5G results in several different configuration permutations [1]. RAN application(s) can be distributed in a flexible way over both physical (PNF) and virtualized/container (V/CNF) infrastructure, as shown in the middle section of Figure 1. How RAN application

parameters are chosen and tuned is partially dependent on the selected distribution over compute, transport and site resources. The right side of Figure 1 shows the four automation domains that play a role in optimal network tuning across applications and infrastructure:

- » DevOps (development and operations) design flow
- » Network design/operations and maintenance (O&M)
- » Dynamic infrastructure optimization
- » RAN-algorithm and user-centric optimization.

Sophisticated use of AI techniques involves deep and disruptive changes to many existing software engineering, algorithm design and workflow stages. The DevOps (Development and Operations) flow, for example, needs to be extended with new data and ML model life-cycle management (MLCM) procedures. Network design/O&M, dynamic infrastructure optimization and RAN-algorithm/ user optimization also need to be reengineered in a radically but carefully staged approach [2].

Software-enabler framework for AI-based RAN automation

Our research and supporting proof-of-concept systems have shown that a set of software enablers is required to best facilitate new AI approaches to sophisticated closed-loop automated network design. These enablers must be tightly integrated into a software framework that allows flexible and open information exchange. As shown on the left side of Figure 1, our software-enabler framework for AI-based RAN automation consists of nine key enablers:

- » A data layer (E1)
- » A message bus (E2)
- » Training and inference (E3)
- » Model life-cycle management (E4)
- » Open-source AI software (E5)
- » Integrated simulation (E6)
- » A decision-making agent layer (E7)
- » Rules and policies for intent-based management (E8)
- » A digital sibling for knowledge harvesting (E9).

Perhaps the most essential enabler of all is the production of industry-standard, machine-readable data (E1) at a high temporal rate by RAN network nodes (eNodeB or gNodeB). This data can be aggregated, stored, filtered and distributed by advanced real-time, hierarchical, message bus (E2) solutions. A distributed training and inference (E3) architecture can efficiently consume this data and use it to facilitate the training of ML models to create new AI algorithms and/or parameter selection for classic algorithm design [2].

There are many alternatives [3] to creating/training models for use with ML algorithms, with three main classifications:

1. Supervised learning, in which algorithms are trained using labeled examples
2. Unsupervised learning, using a cluster or grouping technique
3. Reinforcement learning (RL) [4], using a target reward strategy that allows guidance toward an optimal set of actions.

Depending on the problem targeted, the best ML model and supporting life cycle (E4) can vary considerably in complexity. For simpler problems, linear regression, smaller decision-tree and simple neural networks (NNs) with a few nodes and layers can be sufficient. For more complex problems, large decision trees or deep neural networks (DNNs) with many layers and nodes and several convolutional layers could be necessary to achieve the required accuracy.

RL approaches and supporting agents (E7) for controlling optimization goals can be especially effective at learning novel RAN management strategies. Training RL models depends on active exploration, through a software agent’s trial and error experiments, which will not always be possible or appropriate in a live RAN system. To help solve this problem, and to generate the required quantities of data to train models, we have included simulation (E6) in our set of software enablers.

Once trained, an ML model can be used in the inference phase (part of E3), where a selection of data is used as input into the model that will then produce a set of predictions, actions or rules, with the exact details depending on the ML algorithm type. In RAN, the hardware and software requirements on the training and inference phases can be vastly different. Training typically requires powerful central processing unit or specialized graphics processing unit (GPU) hardware with large memory and data storage.

AI software platforms, such as TensorFlow, Keras and PyTorch, together with other extensive, open source (E5), often Python-based, ML software ecosystems need to be integrated into the software engineering flow. During the inference phase, a trained model (or models) is made available to the RAN application(s) through model life-cycle management (E4). For latency-critical RAN edge applications, the inference needs to be efficiently realized, with low latency, power consumption and memory footprint, taking the characteristics of the target hardware and software architecture into account.

Our software enablers are fully compatible with intent-based management solutions [5].

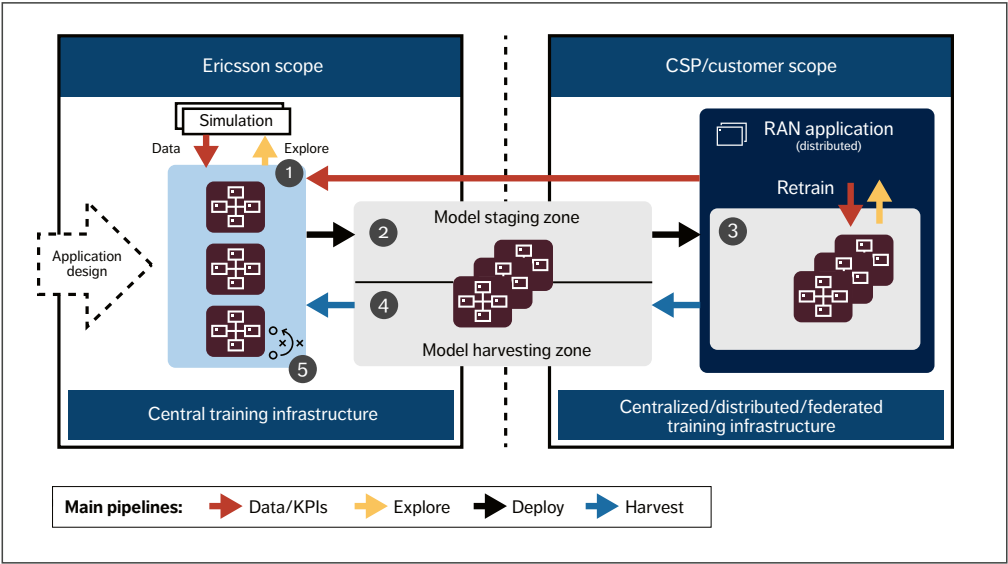


Figure 2 Model life cycle management

When moving toward AI-based automation, operators will no longer spend time tweaking RAN performance by adjusting individual parameters exposed by the RAN system. Instead, well-defined high-level rules and policies (E8) for expressing overall operational intent are used. The AI-based automation machinery is then responsible for realizing these goals.

Long term, to allow the most sophisticated forms of automation, we see a need to build partial digital representations of deployed systems. Our unique digital sibling (E9) concept makes it possible to harvest ML models to build context-specific knowledge.

Model life-cycle management

Machine-learning models are valuable, evolving assets. As such, a sophisticated and controlled software engineering chain (E4) is required for MLCM. When a model is updated through retraining outside its initial context, for example, many significant challenges are introduced in terms of traceability, stability and the possibility to identify

faults. With potentially many instances of a model deployed at cell, cell cluster or indeed user level, this challenge is compounded. Figure 2 captures the main conceptual stages in model management from a software process perspective.

After an initial model design and training phase (marked as 1 in Figure 2), possibly supported by simulation components, the model enters a staging zone (phase 2). This is where customer- or hardware-specific model adaptations [6] occur and are packaged into an industry-deployment format. For example, a DNN can be pruned to minimize nodes that are not significant with additional quantization transformation to reduce the number of bits required to represent a number. This can significantly reduce size, producing an embedded DNN device representation with just a minor impact on accuracy. In phase 3, the model(s) is deployed into a RAN application for use. Initially, these models will be tightly integrated with the RAN application. In the long term, however, it will be necessary to separate model and application life cycles to allow for rapid model training/retraining and deployment cycles.

●● WE ENVISION THE USE OF OUR NOVEL MODEL-HARVESTING TECHNIQUE TO CREATE DIGITAL SIBLINGS ●●

Retraining and model drift

Certain types of models (those dependent on traffic and variation in load, for example) may not initially perform well in their deployed context. Over time, there may also be drift between the model training context and the current operating state. As such, it is necessary to allow for periodic, active retraining with context-specific data and features in phase 3.

Model harvesting

Model harvesting is a way to improve simulation and ML training capabilities by enabling a model feedback loop from deployment to design contexts. Phase 4 includes the anonymized collection of model data from a CSP deployment back into our model harvest zone. This facilitates sophisticated data mining from these models, including valuable clustering information on RAN resource usage. The harvested data can then be used as input for model training and retuning when the simulator configurations are updated, for example. In the longer term we envision the use of our novel model-harvesting technique to create digital siblings (phase 5) that partially reflect different real-world RAN deployments.

Test, verification and run-time monitoring of RAN applications utilizing machine learning

A data-driven and continuously learning RAN will introduce new requirements on how software is tested, verified and monitored during system run time. Today, exhaustive test and verification is performed, both on individual software modules and at system level, automated over various continuous integration loops before deployment. The same software is deployed on most RAN nodes, perhaps somewhat differentiated depending on the hardware, surrounding equipment and customer preferences. However,

when moving toward RAN systems that utilize ML and RL technologies, this will no longer hold.

For example, ML models may develop independently and not be the same for all RAN nodes, and they may not remain constant between RAN software releases. As a result, new ways to continuously test and validate RAN software utilizing ML models must be developed to fit the MLCM phases.

Moreover, new fine-grained KPIs and metrics must be collected during system run time and fed back to the MLCM. This includes new KPIs for deployed ML models, such as their perceived accuracy, as well as performance and resource consumption KPIs for the RAN applications utilizing the ML models. Similar to other type of RAN data relevant for machine learning, the KPIs should be distributed by the data layer.

A data layer targeting machine learning in the RAN

A cornerstone for producing high-quality ML and RL models is good access to data. Currently, there are many mechanisms for the generation, collection and analysis of data from the RAN. One example is the performance management and configuration management data reported in a standardized format from RAN nodes. Another example is debug and trace data, which includes detailed information about the internal RAN system state. The latter data is often generated on demand – that is, associated with a trouble report or customer service requests.

When targeting generation and use of ML models for time-critical RAN control loops, none of the existing mechanisms is a perfect match. At Ericsson, we are actively investigating and introducing new mechanisms for the generation, management and distribution of data suitable for ML in the RAN. Our lightweight data generation mechanisms allow high quantities of data to be generated without adversely affecting core RAN system performance.

The data is produced using a few standardized machine-interpretable formats. Each data format has an associated schema allowing for easy serializing, deserializing and content interpretation. The data

management allows different data sources to be controlled and activated in a very fine-granular manner. Information on the data and associated schemas is stored and organized in distributed databases, called data- and schema inventories.

Our data distribution is based on message bus technologies, allowing the produced data to be automatically tagged with relevant keys and to be automatically routed to different subscribers based on their tag subscription requests. Our research-developed message bus is hierarchical and distributed, meaning that the produced data will not be routed longer than needed, making it particularly suitable for the requirements of a virtualized and disaggregated RAN.

Enhancing machine learning in the RAN with simulation

Despite ongoing work to improve high-fidelity data collection in the RAN, its distributed nature means it is likely that data will continue to be sparse and that data collection will continue to be a resource-consuming task. At the same time, learning through exploration can be difficult in live systems, where exploration may degrade or crash the system.

Our approach to mitigating these challenges is to integrate in-loop simulation (E6) into our framework, providing the means for much richer data generation as well as the use of explorative RL techniques. Allowing in-loop simulation creates new possibilities for complex, multistep, scenario exploration without compromising system integrity. The first phase of our in-loop simulation is restricted to the Ericsson development scope shown in Figure 2, but it will evolve to include customer deployment in the future.

Looking ahead, more sophisticated AI-based automation in the RAN will require partial digital representations of deployed systems to support complex decision-making. In our framework, we call this enabler a digital sibling (E9).

Digital sibling

The digital sibling is a way to capture and transfer data, and especially ML model data, from a real-world context. These digital representations can then be used for increasingly accurate data generation and in-loop simulation. A related concept, the digital twin, endeavors to duplicate, more completely, a full digital copy of a physical resource or system. The scope of our digital sibling is limited to the transferable digital knowledge, as captured in models that have been trained, interchangeably, in both a simulated and real-world RAN environment.

After an initial training phase, an ML model captures a considerable amount of transferable, problem-specific knowledge, including knowledge that is compressed and coded in the weights of an NN or a set of rules in a decision tree. Sets of models can be used to capture the different decision-making needs of a system. For example, a set of ML models can be trained in a local CSP context to predict the best link adaption or power control parameters for a user.

Other examples include the choice of geographical baseband workload placement required for a given service category and anomaly detection like rogue users or a base station with malicious intent. This aggregated model information (transferred through our MLCM in phase 4), together with actions by goal-oriented software agents, forms a considerable body of knowledge that

Terms and abbreviations

AI – Artificial Intelligence | CSP – Communication Service Provider | DevOps – Development and Operations | DNN – Deep Neural Network | DSP – Digital Signal Processor | GPU – Graphics Processing Unit | ML – Machine Learning | MLCM – Model Life-Cycle Management | NN – Neural Network | PNF – Physical Network Function | RL – Reinforcement Learning | V/CNF – Virtualized/Container Network Function

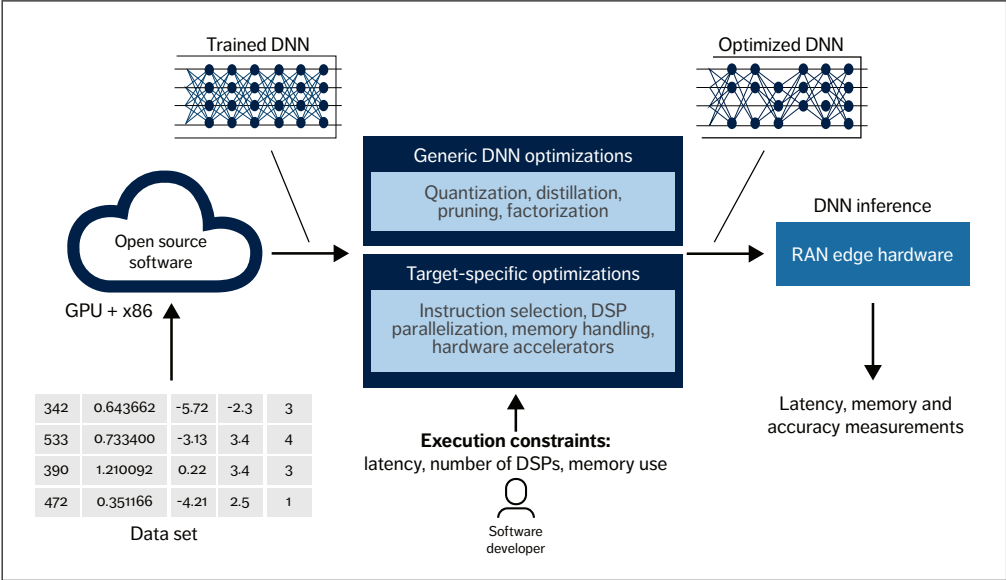


Figure 3 DNN inference optimization for RAN edge hardware

can continuously evolve to form a learning system. The digital sibling is a unique part of our framework, which will ultimately enable advanced automation.

Proof of concept #1 – deep neural network inference optimization

In the proof of concept shown in *Figure 3*, we have investigated how to tune, optimize and realize pretrained DNNs for inference on RAN edge hardware. We selected this example because radio- and baseband data processing are time-critical and computationally intensive tasks. These steps would be part of phase 2 in *Figure 2*.

The DNN training was done in a container-based environment (Kubernetes) using GPUs and state-of-the-art open source DNN training frameworks. Several generic DNN optimization techniques were investigated and implemented, such as quantization, distillation and pruning, with the goal to reduce the footprint and execution time of the DNN inference, still keeping the accuracy close to the accuracy of the originally trained DNN.

The optimized DNN was realized in terms of digital signal processor-C (DSP-C) code – that is, C extended with some specific DSP keywords and instructions, taking into account constraints given by the target RAN hardware and the RAN software designers, such as the number of DSPs and the amount of memory available for the DNN inference. Latency and memory measurements were made on the RAN edge hardware platform for the generated DNN DSP-C code. We also tuned how to best parallelize the DNN inference over several DSPs.

Based on performed evaluations, we concluded that DNN inference for the target hardware can benefit significantly from the generic- and target-specific DNN optimizations implemented. However, there is normally a trade-off between ML model accuracy, execution time and memory consumption.

Proof of concept #2 – AI-accelerated RAN platform

The proof-of-concept platform shown in *Figure 4*

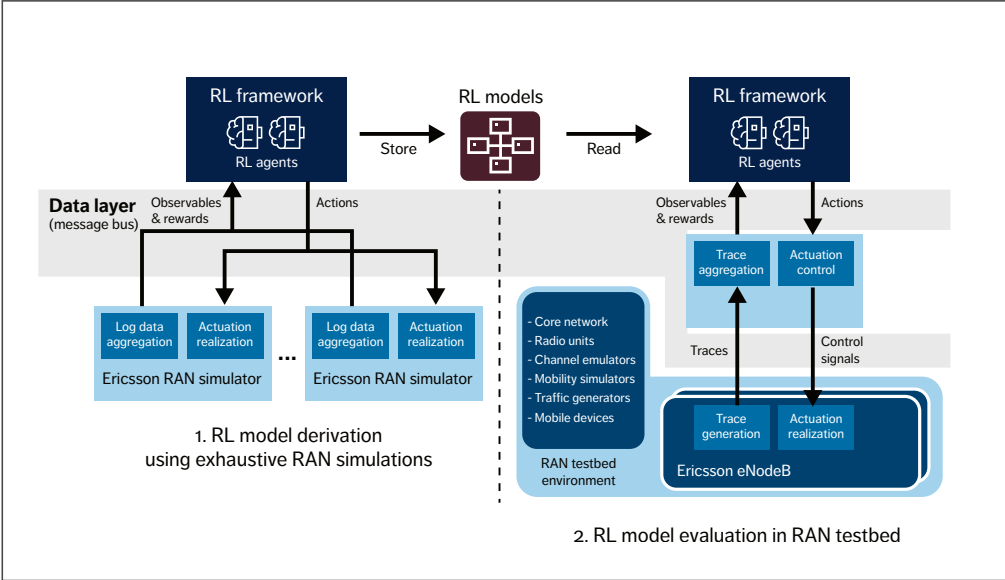


Figure 4 AI-accelerated RAN platform

supports the use of RL techniques and model transfer to control parts of the RAN functionality. Our second proof of concept explores phases 1, 2 and 3 in *Figure 2*. We have used sophisticated RAN simulators to set up and run many different deployment scenarios to train the RL models. The RAN test bed we used consists of several cells and multiple UEs (user equipment). The data layer uses a message bus, which organizes, aggregates and filters data.

To obtain a high degree of scalability and improve model generalization and RL exploration possibilities, the RAN simulators have been containerized, allowing many concurrent instances to execute with different configurations.

The RAN simulator can extract observables and rewards based on internal simulator data, as well as change its internal state and resource allocation strategies based on the actions provided by the RL agents. The RL agent framework is separated from both simulated and actual RAN application by the same well-defined data and action interfaces. RL algorithms can be developed and evaluated in

isolation from the RAN application, using RL domain experts, and with required compute and memory resources, including the use of powerful GPUs to train the DNNs that are used by the RL agents.

After achieving high performance in the different simulator scenarios, the simulator-trained RL models can be used for automation use cases in real-life RAN contexts.

Conclusion

Growing complexity and the need to solve repetitive tasks in 5G and future radio systems necessitate new automation solutions that take advantage of state-of-the-art artificial intelligence and machine learning techniques that boost system efficiency. At Ericsson, we have identified the key software enablers for AI-based RAN automation and integrated them into a comprehensive framework that provides a solid and flexible technological foundation for the development of AI-based RAN. Our two proof-of-concept examples, executed in real RAN contexts, clearly illustrate the benefits of our framework of software enablers in building the next generation of automated RAN systems.

References

1. Ericsson Technology Review, 5G New Radio RAN and transport choices that minimize TCO, November 7, 2019, Eriksson, A-C; Forsman, M; Ronkainen, H; Willars, P; Östberg, C., available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/5g-nr-ran-and-transport-choices-that-minimize-tco>

2. Ericsson Technology Review, Enhancing RAN performance with AI, January 20, 2020, Calabrese, F.D.; Frank, P; Ghadimi, E; Challita, U; Soldati, P., available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/enhancing-ran-performance-with-ai>

3. MIT Press, Deep Learning, 2016, Goodfellow, I; Bengio, Y; Courville, A., available at: <https://www.deeplearningbook.org/>

4. MIT Press, Reinforcement Learning: An Introduction (second edition), 2018, Sutton, S; Barto, A., available at: <http://incompleteideas.net/book/RLbook2020.pdf>

5. Ericsson Technology Review, Cognitive technologies in network and business automation, June 28, 2018, Niemöller, J; Mokrushin, L., available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/cognitive-technologies-in-network-and-business-automation>

6. KTH Royal Institute of Technology, Squeezing and Accelerating Neural Networks on Resource Constrained Hardware for Real Time Inference, 2018, Presutto, M., available at: <http://www.diva-portal.org/smash/get/diva2:1299467/FULLTEXT02.pdf>

Further reading

» Ericsson, How will AI enable the switch to 5G?, available at: <https://www.ericsson.com/en/networks/offerings/network-services/ai-report>

» Ericsson, Autonomous networks, available at: <https://www.ericsson.com/en/future-technologies/autonomous-networks>

» Ericsson, Ericsson launches unique AI functionality to boost radio access networks, available at: <https://www.ericsson.com/en/news/2019/10/ericsson-ai-to-boost-ran>

» Ericsson, AI by Design, available at: <https://www.ericsson.com/en/ai-and-automation>

» DeepMind blog, AlphaGo Zero: Starting from scratch, available at: <https://deepmind.com/blog/article/alphago-zero-starting-scratch>

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5G synchronization requirements and solutions

The need for synchronization in the RAN has grown as new radio technologies and network architectures emerge to boost efficiency and support demanding 5G use cases. Although the fundamental synchronization requirements have not become more stringent in 5G, the need for time synchronization has become much more critical.

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Accurate and reliable synchronization has long been a fundamental prerequisite for the correct operation of telecommunications networks. Its importance has grown in 4G, and it will be more important than ever in 5G and future networks.

■ RANs are designed to optimize service performance and reliability, and synchronization is a vital enabler. The RANs use different sources, often in combination, to provide synchronization characteristics to match the network use case and services delivered. Finding the right balance between timing accuracy, availability and cost is key to making services successful.

Many of the commercial 5G networks going live around the world today use TDD. TDD radio frames inherently require time and phase alignment between radio base stations, to prevent interferences and related loss of traffic. Time synchronization is also required in FDD networks when different radio coordination features are used.

New network architectures imply new approaches to addressing the various synchronization needs. This includes using Precision Time Protocol (PTP) and radio interface-based methods to synchronize distributed radio units in the evolved RAN architecture, where the upper and lower parts of the 5G New Radio (NR) RAN are separated in the different logical units: the centralized unit (CU),

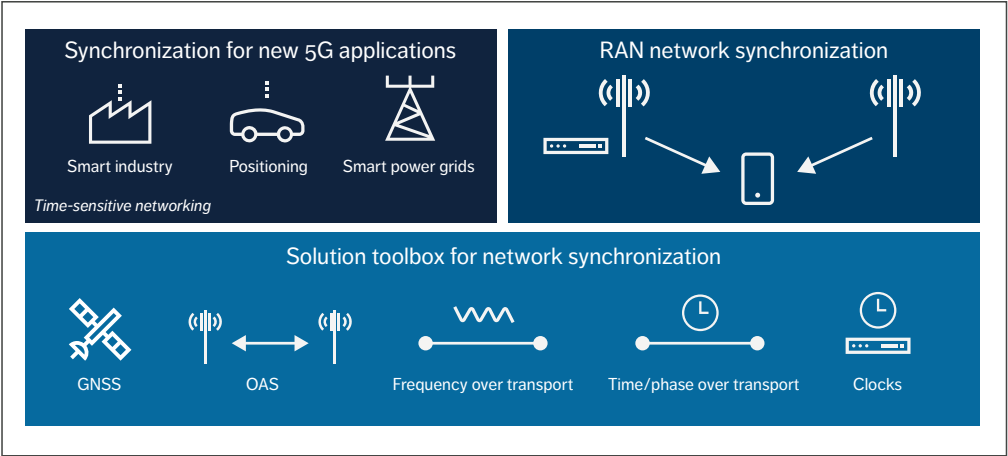


Figure 1 Synchronization in new network scenarios

the distributed unit (DU) and the radio unit (RU). The baseband function in a base station is split into two logical units: a CU hosting the higher-layer protocols and a DU handling the lower layers to the user equipment (UE).

5G makes it possible to serve several applications that have previously been supported either by wireline or by non-standard radio technologies, such as Critical Internet of Things (IoT) and Industrial Automation IoT services [1] for which time synchronization is fundamental. While many applications benefit from accurate time synchronization, it is important to realize that high time accuracy over large areas can be very costly. Services with special accuracy or availability demands should be carefully analyzed to minimize costs.

Figure 1 provides some examples of synchronization in new network scenarios. Time and phase synchronization is particularly important in 5G networks.

The two main types of synchronization requirements that are relevant for 5G networks are those that depend on the radio network operation and those that depend on the supported services (application-driven requirements).

Radio network-driven synchronization requirements

Synchronization requirements related to communication over the radio interface can be divided into two categories: TDD cell phase synchronization (Tsync) and communication features based on coordinated transmission or reception from multiple Transmission Reception Points (TRxPs).

TDD cell phase synchronization

TDD cells operating at the same frequency (or adjacent frequencies) in overlapping coverage areas require time domain isolation to prevent base-station-to-base-station and UE-to-UE radio frequency (RF) interference.

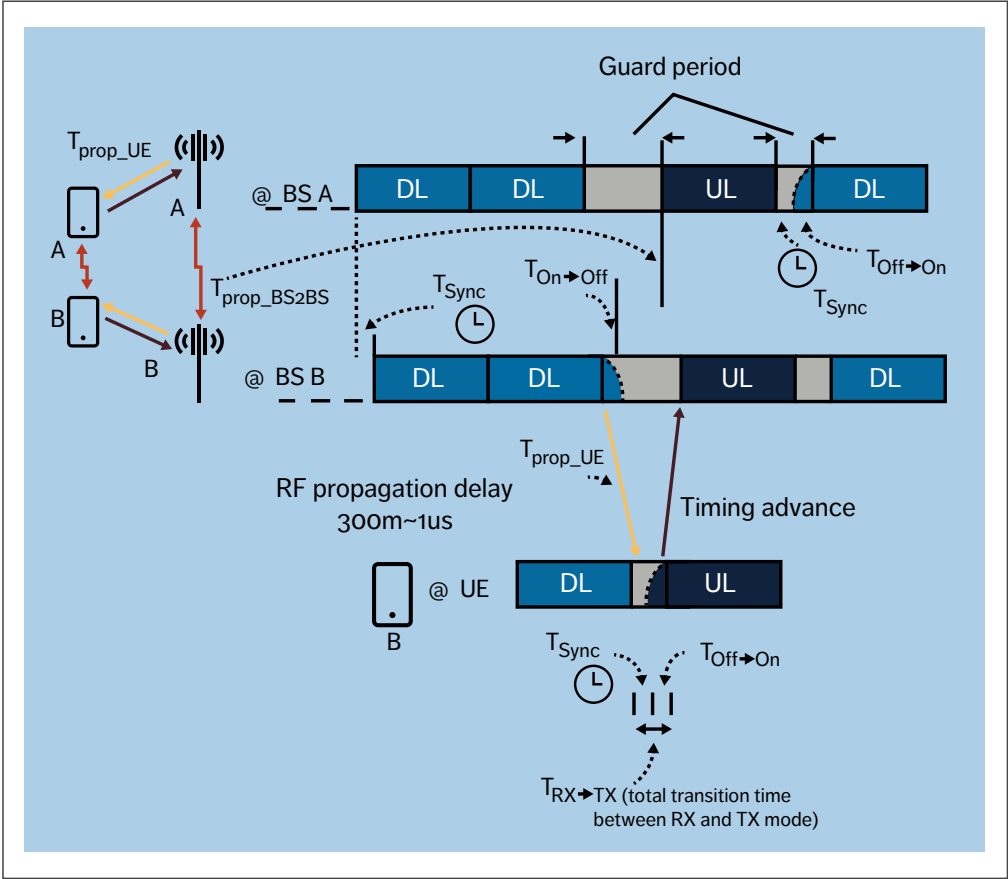


Figure 2 Synchronization for TDD

There are two requirements for time domain isolation:

- » the cells must use the same TDD configuration
- » the deviation in frame start timing between cells must be below a maximum value specified as cell phase synchronization accuracy in the 3GPP.

For TDD synchronicity and interference, the critical points are when switching between transmission and reception, as shown in Figure 2. Guard periods are used for isolation with a configurable total guard time expressed as an integer number of symbols. The duration of the guard period needs to cater for four effects:

- » air propagation time (T_{prop})
- » sufficient transient time when the transmitter changes between defined ON/OFF power levels ($T_{On \rightarrow Off}$, $T_{Off \rightarrow On}$)
- » sufficient time for the UE and base station to change between transmit and receive modes ($T_{RX \rightarrow TX}$, $T_X \rightarrow T_{RX}$)
- » allocate margin for cell phase synchronization errors (T_{sync}).

Since guard periods are not used for communication, they reduce the time ratio when spectrum resources

can be used. To decrease overheads, the emphasis is on striving to keep the guard periods short, while still catering for the desired effects. Allocation of a budget to the different components of the guard period is the result of a trade-off between cost (product and deployment), availability, TDD periodicity and overheads.

In 3GPP New Radio (NR), cell phase synchronization is specified as 3µs [2] – that is, the same as for LTE. This is because the reduced transient times in NR made it possible to keep the same synchronization requirement with low overhead.

The requirement on cell phase synchronization is ultimately specified in terms of maximum deviation in relation to a common absolute timing requirement and dividing the requirement by half ($\pm 1.5\mu s$, for example). This allows a design with independent synchronization references.

In cases where the TDD-unicast area is not isolated, the 3GPP has specified that traceability to a standard timing reference (UTC) is required [3]. This prevents interference between different networks using adjacent frequency bands or between national borders, because it enables the phase alignment of the radio frames generated by different networks with overlapping areas.

UTC traceability does not imply that UTC time is recovered or that leap seconds are used; in fact, the 3GPP also requires the use of continuous time without leap seconds. In practice, a base station with a Global Navigation Satellite System (GNSS) receiver fulfils the requirement on UTC traceability as specified by the 3GPP. The ITU-T (International Telecommunication Union Telecommunication Standardization Sector) specification of the primary reference time clock (PRTC) includes the possibility for the PRTC to be synchronized with GNSS. This means that a PTP-based solution can also fulfill this requirement.

THIS ALLOWS A DESIGN WITH INDEPENDENT SYNCHRONIZATION REFERENCES

Communication features based on coordinated transmission or reception from multiple Transmission Reception Points

A variety of features that benefit from coordinated transmission or receptions from multiple TRxPs have been standardized over the years, all with different purposes and characteristics. Some relate to combining spectrum assets, thereby allowing total higher aggregated bandwidth and throughput (carrier aggregation, dual connectivity and so on), while others relate to improving link performance at the cell edge (variants of coordinated multi-point operation, for example). Still others relate to specific services like Multimedia Broadcast Multicast Service over a Single Frequency Network. Coordination may also occur between NR and LTE.

These features and related timing requirements are applicable within a single operator network, and as a result, control of relative time error between antennas used by the feature is sufficient. What matters is the relative frame timing alignment at the receiver. The 3GPP has therefore defined a maximum receive timing difference (MRTD) as a maximum relative receive timing difference the UE must be capable of handling. The MRTD consists of a base-station relative time alignment error (TAE) and an RF propagation delay difference (ΔT_{prop}). That is, $MRTD = TAE + \Delta T_{prop}$.

Based on the level of required MRTD, three main categories could be identified:

- » MRTD as a fraction of a cyclic prefix (CP)
- » MRTD without CP relation
- » no timing requirement.

In the case of MRTD as a fraction of the CP, the remaining duration of the CP allows for channel delay spread. In 3GPP TS 38.104 [10], the TAE ranges between 65ns to 260ns depending on the feature and on the CP duration but is only valid for colocated/ intra-site deployments where $\Delta T_{prop} \sim 0$. The strict MRTD, and thereby TAE, relates to features using the same frequencies (like MIMO (multiple-input, multiple-output)) or at adjacent spectrum (like contiguous carrier aggregation) where common or shared functions in the RF chain could cause strict timing dependencies.

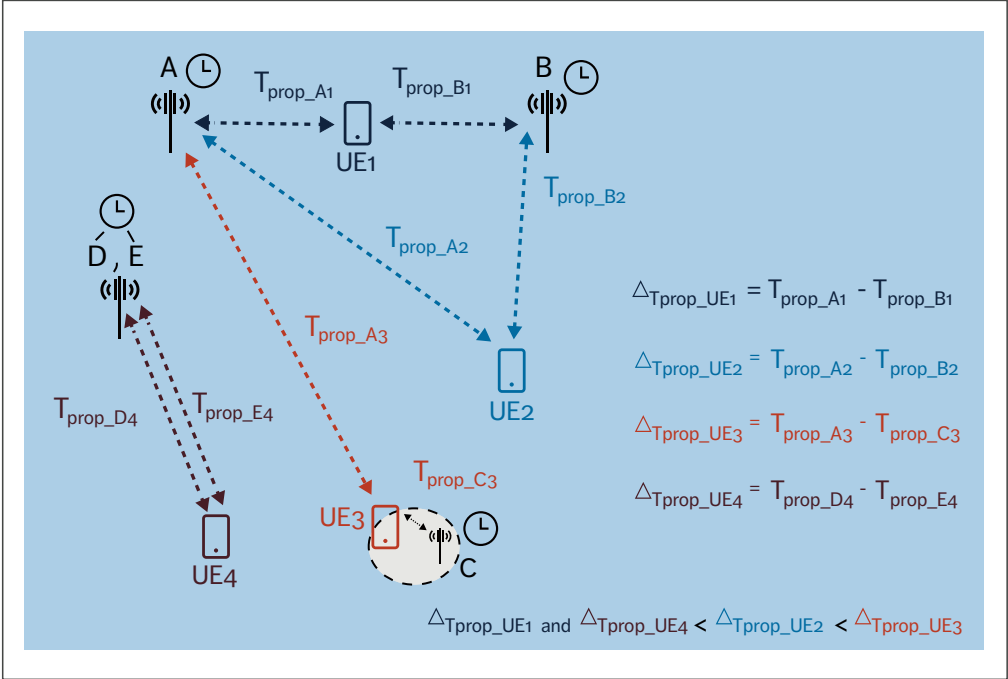


Figure 3 Synchronization for coordinated transmission/reception

Even if features in this category can be served from distributed TRxPs (where generally $\Delta T_{prop} \neq 0$) – that is, not being restricted to a colocated/inter-site deployment scenario – it would be difficult to specify and mandate a single fixed TAE. Rather, the required TAE would depend on actual deployment, its RF characteristics and the UEs’ relative position between the TRxPs, as shown in Figure 3.

As noted in the bottom of the figure, ΔT_{prop_UE1} and ΔT_{prop_UE4} are less than ΔT_{prop_UE2} , which is less than ΔT_{prop_UE3} . For the same delay spread, UE1 and UE4 can tolerate larger TAE than UE2 and UE3. For colocated D and E, the TAE of D-E is generally less than the TAE of A-B.

In contrast, features in the “MRTD without CP relation” category allow a relatively large part for ΔT_{prop} and thereby more flexible deployments, such as heterogeneous networks.

For example, synchronous dual connectivity allows an MRTD of up to 33µs, of which 30µs can be allocated for ΔT_{prop} and the remaining 3µs for TAE. 3GPP-defined asynchronous dual connectivity without TAE requirements and ΔT_{prop} restrictions is an example of a feature with no timing requirement.

Application-driven synchronization requirements
While the introduction of 5G did not cause any fundamental change to radio network synchronization requirements, some applications may put more stringent local accuracy requirements on the synchronization of the 5G nodes. Examples include time-sensitive networks (TSNs), smart grid applications and the UE device-positioning use case.

OAS METHODS PROVIDE AN ATTRACTIVE SYNCHRONIZATION ALTERNATIVE

Time-sensitive networks
TSNs in industrial applications generally require the distribution of time synchronization to the industrial subnetworks used for functionality such as robot control or autonomous vehicles. This results in synchronization requirements in the microsecond range between nodes in the TSN. The 3GPP technical specification TS 23.501 [4] presents an architecture and methods that enable the successful integration of a 5G network into a TSN synchronization network. Industrial automation scenarios often involve multiple timing domains. The recommended solution is to relay timing transparently across the 5G network, where the delay of the TSN-time messages through the 5G network is measured and compensated. The same architecture can also be used to support the industrial application by distributing the timing of the 5G network to the TSN [5].

Smart grid applications
When 5G is used to support smart grid applications, the 3GPP synchronization architecture [4] can be used to support the distribution of timing to the synchrophasors that in this way can measure the phase relationships in the AC power distribution network. These applications require synchronization accuracy in the same range as the TSN. In this case, the distribution of a single time reference is sufficient.

Positioning of UE devices
Some approaches to solving the UE device-positioning use case rely on accurate synchronization of the radio nodes with the help of methods such as OTDOA (Observed Time Difference of Arrival). Depending on the required positioning accuracy, these may not always result in

cost-efficient solutions. For example, 1m corresponds to a synchronization accuracy of around 3ns, which is orders of magnitude more stringent than other radio network synchronization requirements. A method that depends on such a high level of synchronization accuracy would significantly increase the cost of the solution, especially when synchronization to the base stations is distributed over the network. Over-the-air synchronization (OAS) methods provide an attractive synchronization alternative in this case.

Evolving network synchronization standards and technologies
Most of the relevant synchronization requirements are defined by the 3GPP. The fundamental technologies to address these requirements come from the GNSS, which provides the basic technology for a time synchronization master, enabling global distribution of a UTC-traceable reference. GNSS services include GPS, BDS (the BeiDou Navigation Satellite System) and Galileo [6], and are offered by several space agencies. GNSS is based on satellites having a known time and orbit position, transmitting messages whose arrival are measured by the GNSS receiver. GNSS satellites visibility is important for proper operation.

Specified by IEEE 1588, PTP has been selected by the telecom industry to distribute accurate time synchronization from an accurate master, such as GNSS. The basic concept is to distribute time synchronization from the PTP “grandmaster” to the leaf PTP clocks, using PTP messages. The protocol is based on a two-way exchange of timing messages [7]. This message exchange enables the distribution of the time from the PTP grandmaster and estimates the path delay. The path delay estimation assumes that the delay of the two transmission directions is symmetric. This is not always the case in real networks due to factors such as different fiber length or the use of different wavelengths in the optical transmission. Since any source of asymmetry results in error in the time synchronization distribution, the asymmetry must either be known and compensated for or sufficiently small to be negligible.

To limit the effects of asymmetry and packet delay variation that could be created by the network nodes (switches and routers, for example), IEEE 1588 has specified the “boundary clock” and “transparent clock” functions with event message timestamping in the hardware layer.

The distribution of time-phase and frequency synchronization has been addressed in several ITU-T recommendations [8]. In particular, these recommendations define reference synchronization networks, where the synchronization is generated by time synchronization masters or PRTCs, which are typically based on GNSS technology and where the reference timing signal is carried across a network of clocks. Frequency synchronization in these reference networks is carried over the physical layer (typically using synchronous Ethernet). The time synchronization reference is carried through a PTP, where the PRTC is the source of time for the PTP Grandmaster, also known as the Telecom Grandmaster (T-GM).

●● IN MANY CASES, A COMBINATION OF TECHNIQUES IN BOTH DOMAINS WILL BE REQUIRED ●●

Two PTP profiles have been defined for the use of PTP in telecom – G.8275.1 (PTP with full timing support from the network) and G.8275.2 (PTP with partial timing support from the network) [8]. The purpose of the G.8275.1 profile is to meet the highest accuracy, which requires the implementation of an IEEE 1588 telecom boundary clock or telecom transparent clock in every node in the timing distribution network. Assisted partial timing support (APTS), which concerns the use of PTP as a backup for GNSS, is an important implementation of the G.8275.2 profile.

To meet the TDD “relative” synchronization requirement that cell phase synchronization must be better than 3µs, the accuracy of the time synchronization reference at the input to the base

station must always be better than ±1.1µs when measured against a standard time reference. A margin of 400ns is left to the end application to fulfill ± 1.5µs on the radio interface. Synchronization methodologies and architectures applicable to a switched fronthaul have also been covered in G.8271.1 [8].

The synchronization aspects of fronthaul networks are also addressed by the O-RAN Alliance. In particular, one of the groups in O-RAN addresses the split between the DU and RU – O-DU and O-RU in the O-RAN terminology. The split option is denominated as 7-2x and it is an intra-PHY split – that is, some parts of the 3GPP PHY layer are implemented in the DU and others in the RU [9].

To address the synchronization needs in these scenarios, the O-RAN specification defines different synchronization topologies. The DU may be part of the synchronization chain or just a consumer of the synchronization. The RU is the main user of synchronization and it may also be synchronized by a timing reference generated directly at the radio site. The data carried between a DU and an RU uses Ethernet-based transport. The time and frequency synchronization distribution can be achieved using a combination of PTP and synchronous Ethernet. There are no specific synchronization requirements for the CU.

Recommended synchronization solutions for evolving mobile networks

We have compiled a comprehensive toolbox of synchronization techniques that can be used to provide appropriate frequency synchronization and time alignment of the radio network with minimal disruption. Some of them are implemented in the RAN domain, while others are implemented in the transport domain. In many cases, a combination of techniques in both domains will be required to create a robust and reliable solution.

RAN-based synchronization solutions

A solution is considered to be RAN based if it can fulfill the synchronization requirements of the RAN

network without synchronization support from the transport network. The two main RAN-based solutions are GNSS-based synchronization and over-the-air synchronization (OAS).

A GNSS-based solution installed directly at base station sites can provide cost-efficient, accurate and predictable time synchronization of the radio network without any support from the transport network. The higher accuracy of GNSS compared with other synchronization sources allows for comparatively longer holdover periods. Moreover, the time error can be better estimated during holdover, making it possible to take consequent actions per service (TDD, for example).

●● TRANSPORT-BASED SOLUTIONS ... RELY ON TWO KEY TECHNOLOGIES ●●

Short disruptions due to interference and blocking are common, but the negative impact can be avoided or limited by the base station’s holdover using the local oscillator – up to a few hours for a TDD base station, for example. Longer holdover periods can be achieved when GNSS is combined with other technologies that provide a stable timing reference (traceable to PRTC).

One promising method for time alignment of the radio network is to use information over the NR air interface. OAS uses timing information sent over the radio interface between neighboring base stations. Features requiring stricter synchronization are generally implemented between neighboring base stations. OAS is based on round-trip time measurement similar to PTP, with the advantage that the asymmetries are generally small over the air, making it possible to achieve good time alignment.

Transport-based synchronization solutions

Transport-based solutions, in which synchronization is distributed over the transport network, rely on two key technologies: frequency synchronization over the physical layer (synchronous Ethernet) and frequency and time synchronization over the packet layer (PTP).

Frequency synchronization based on packet protocols like Network Time Protocol (NTP) or PTP using the G.8265.1 profile (in a PTP-unaware network) has proven to be the most cost-efficient solution in LTE-FDD networks, where frequency synchronization alone was often sufficient. In 5G networks the situation is different because time and phase synchronization is more often needed compared with previous generations of radio networks, due to factors such as wider use of TDD or the need for coordination between NR and LTE.

Terms and abbreviations

APTS – Assisted Partial Timing Support | CP – Cyclic Prefix | CPRI – Common Public Radio Interface | CU – Centralized Unit | DL – Downlink | DU – Distributed Unit | eCPRI – Enhanced CPRI | ePRTC – Enhanced Primary Reference Time Clock | FTS – Full Timing Support | GNSS – Global Navigation Satellite System | IoT – Internet of Things | ITU-T – International Telecommunication Union Telecommunication Standardization Sector | MRTD – Maximum Receive Timing Difference | NR – New Radio | NTP – Network Time Protocol | OAS – Over-the-Air Synchronization | PRTC – Primary Reference Time Clock | PTP – Precision Time Protocol | RF – Radio Frequency | RU – Radio Unit | RX – Reception | T-GM – Telecom Grandmaster | TAE – Time Alignment Error | TSN – Time-Sensitive Network/Networking | TX – Transmission | UE – User Equipment | UL – Uplink | UTC – Coordinated Universal Time

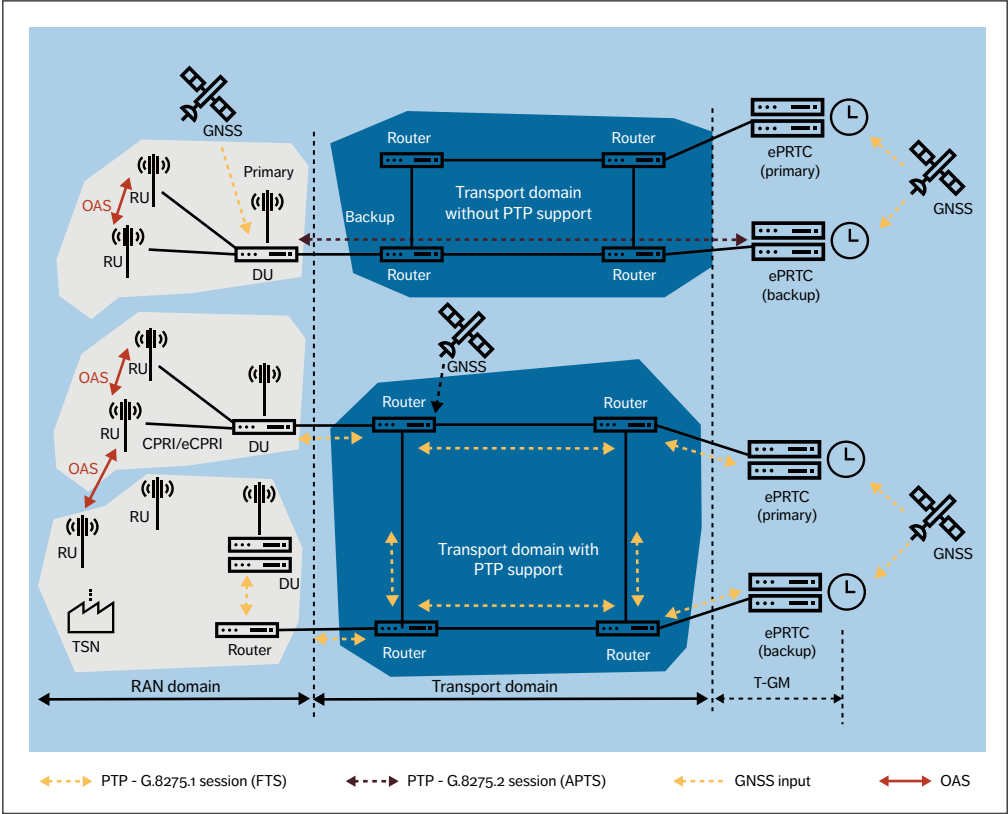


Figure 4 Recommended synchronization solutions

In these cases, the delay asymmetries and packet delay variation present in PTP-unaware networks would make it impossible to meet the stringent accuracy and stability requirements. To mitigate such effects, the transport network requires PTP-aware network elements such as boundary clocks or transparent clocks as well as support of PTP across the entire time distribution chain (“full timing support from the network” as described in G.8275.1). An important aspect of this is that a base station may use PTP for the delivery of both time synchronization and frequency synchronization.

Distributing time synchronization over the same transport network infrastructure used for user data

has the benefits of providing the same level of robustness and redundancy for timing as for the user traffic itself. Another important aspect of the PTP network planning is time error budget planning, which depends on the dimension of the network and the accuracy of the PTP-aware network elements. There are guidelines for this in G.8271.1 [8].

Combining techniques for best results

Several aspects need to be considered when selecting the most appropriate synchronization solution(s), including installation and operation costs, synchronization accuracy, robustness and availability targets. The optimal solution for a

specific network may depend on existing synchronization feature support of the network and network elements, transport network modernization plans that improve synchronization support, as well as operator preferences. Possible future regulations of national authorities may also be relevant.

In many cases, synchronization in a mobile network will benefit from the implementation of a combination of methods. As shown in Figure 4, a fairly typical 5G mobile network can combine methods like GNSS and OAS at base station sites, time and frequency distribution via the transport network, and redundancy and holdover functions in the nodes. This is particularly relevant for cases that require GNSS redundancy (to protect against jamming or spoofing events, for example), where the distribution of timing over the transport network as a complement to a local GNSS receiver should be considered.

PTP with “full timing support from the network” should be used to meet the relevant time synchronization requirements. Redundancy is achieved through a PTP network that is fed from geographically redundant T-GMs and distributes timing over the same physically redundant topologies that are used for user traffic. The best control over the synchronization network architecture will be achieved by rolling out PTP from the lower part of the transport network, with GNSS receivers and T-GM functionality in base stations or routers, and be extended to include T-GMs that are higher up in the network when feasible.

In parts of the network where transport does not support PTP, PTP should not be used as the primary time source for RAN nodes due to the limitations on the achievable performance. In these cases, it should only be used as an assisting reference in APTS mode for GNSS. In these parts of the network, OAS provides an alternative solution for synchronizing the radio nodes without requiring the support of a transport network.

When the GNSS reference is lost, the holdover time can be prolonged by using assisting synchronization sources such as APTS, OAS

and PRTC/enhanced PRTC traceable synchronous Ethernet. The time period a clock can remain within the required accuracy depends on the stability of the assisting source.

THE TIGHTEST REQUIREMENTS SHOULD BE ADDRESSED LOCALLY ONLY WHEN AND WHERE REQUIRED

Conclusion

Proper network synchronization is a prerequisite to excellent radio network performance. Some of the most compelling use cases for 5G, including industrial automation, depend on more accurate timing and will likely generate additional synchronization requirements in the near future. While the level of the required synchronization accuracy depends on several factors, it would be a mistake to apply the tightest synchronization requirement as a general 5G requirement, as doing so would make the cost for 5G and the future evolution of the mobile technologies unsustainable. The tightest requirements should be addressed locally only when and where required.

An efficient synchronization solution that addresses all relevant scenarios requires the support of a toolbox of synchronization methods that are implemented in the RAN domain (GNSS and OAS), the transport domain (such as PTP), or both. For many features, what matters more than anything else is the relative time error between neighboring base stations. In this context, over-the-air synchronization (OAS) is a powerful tool.

References

1. Ericsson Technology Review, Critical IoT connectivity: Ideal for time-critical communications, June 2, 2020, Alriksson, F; Boström, L; Sachs, J; Wang, Y.-P. E; Zaidi, Ali, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/critical-iot-connectivity>
2. 3GPP technical specification TS 38.133, NR; Requirements for support of radio resource management, available at: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3204>
3. 3GPP technical specification TS 38.401, NG-RAN; Architecture Description, available at: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3219>
4. 3GPP technical specification TS 23.501, System architecture for the 5G System (5GS), available at: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3144>
5. Ericsson Technology Review, 5G-TSN integration meets networking requirements for industrial automation, August 27, 2019, Farkas, J; Varga, B; Miklós, G; Sachs, J, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/5g-tsn-integration-for-industrial-automation>
6. ITU Technical Report TP-GSTR-GNSS – Considerations on the use of GNSS as a primary time reference in telecommunications, 2020, available at: <https://www.itu.int/pub/T-TUT-HOME-2020>
7. IEEE 1588-2019 – IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems, June 16, 2020, available at: <https://standards.ieee.org/standard/1588-2019.html>
8. ITU-T Recommendations G.826x and G.827x series (G.8200-G.8299: Synchronization, quality and availability targets, available at: https://www.itu.int/ITU-T/recommendations/index_sg.aspx?sg=15
9. 3GPP technical specification TR 38.801, Technical Specification Group Radio Access Network; Study on new radio access technology: Radio access architecture and interfaces, available at: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3056>
10. 3GPP technical specification TS 38.104, NR; Base Station (BS) radio transmission and reception, available at: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3202>

Further reading

- » Ericsson white paper, 5G wireless access: an overview, available at: <https://www.ericsson.com/en/reports-and-papers/white-papers/5g-wireless-access-an-overview>
- » Ericsson Technology Review, Integrated access and backhaul – a new type of wireless backhaul in 5G, June 23, 2020, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/introducing-integrated-access-and-backhaul>
- » Ericsson white paper, Cellular IoT in the 5G era, available at: <https://www.ericsson.com/en/reports-and-papers/white-papers/cellular-iot-in-the-5g-era>

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Applied network slicing scenarios in 5G

Network slicing enables new business opportunities across a wide range of use cases and sectors by making it possible to create fit-for-purpose virtual networks with varying degrees of independence. However, the diversity of new commercial and technical requirements has significant implications on how networks are built and managed.

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Private 5G networks – the new business offerings that network slicing enables – deliver functionality that extends well beyond that of the current offerings that are typically based on existing public network services. For example, network slicing makes it possible to create a private 5G network with specific service characteristics as well as varying degrees of security/isolation, exposure, self-management, and so on.

■ There are three main approaches to offering and delivering private 5G networks: the standalone approach, the virtual approach and the hybrid approach. Network slicing enables the customization of system behavior and the isolation of resources/functions for specific services in all three of them. Standalone private 5G networks are independent, on-premises deployments that have limited interoperability with public networks. They may be sold through a mobile network operator, managed by

a customer or provided as a managed service. Network slicing is used to customize the behavior for different use cases/traffic types and to provide isolation between them. A good example of a standalone private 5G network is a dedicated solution deployed at a customer premises such as a manufacturing plant or airport. Although these networks use 5G technologies, they are fully independent and isolated from the public 5G infrastructure.

Virtual private 5G networks, on the other hand, are provided on top of an infrastructure layer that is shared with public services. Network slicing is used to meet customization and isolation needs per use case/traffic type as well as per enterprise customer. Public-safety and connected-car services that make use of the public 5G infrastructure are examples of virtual private 5G networks.

Hybrid private 5G networks are provided by combining infrastructure adopted for public services with infrastructure at a customer’s premises. In

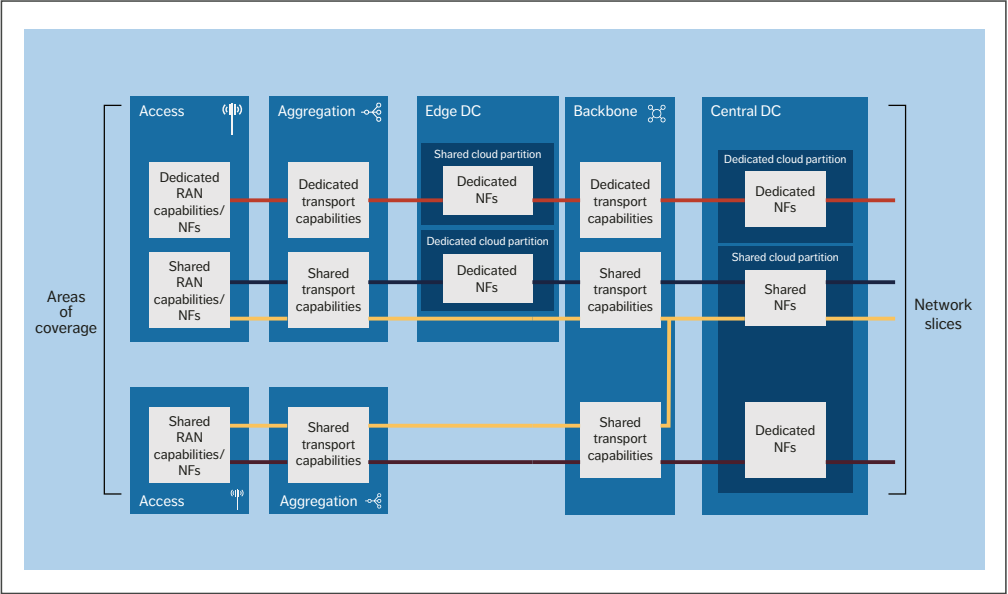


Figure 1 Deployment overview

these networks, network slicing is used to customize and isolate slices per enterprise customer and use case/traffic type. The hybrid approach enables a more flexible distribution of functionality, more efficient use of infrastructure and improved mobility in and out of the customer premises. A good example of a hybrid private 5G network is a manufacturing plant or airport where dedicated on-premises hardware is integrated with and reuses the public infrastructure to improve service and cost-efficiency.

The enablers of network slicing

Figure 1 provides a deployment overview that illustrates the different ways of composing private 5G networks (network slices). To provide the best level of isolation, resources assigned to a network slice are ideally dedicated. Assuming that it is acceptable, some slices may share resources to reduce cost. Distribution and coverage are considered per slice. Some slices are local, while others may be wider in reach. Some slices require

Terms and abbreviations

5GC – 5G Core | **AMF** – Access and Mobility Management Function | **BSS** – Business Support Systems | **DC** – Data Center | **DNN** – Data Network Name | **DRB** – Dedicated Radio Bearer | **E2E** – End-to-End | **EPC** – Evolved Packet Core | **MBB** – Mobile Broadband | **NF** – Network Function | **OSS** – Operations Support Systems | **PDU** – Protocol Data Unit | **RRM** – Radio Resource Management | **RRP** – Radio Resource Partitioning | **SLA** – Service Level Agreement | **SMF** – Session Management Function | **S-NSSAI** – Single Network Slice Selection Assistance Information | **UP** – User Plane | **UPF** – UP Function | **WAN** – Wide Area Network

●● DIFFERENT COMBINATIONS OF ENABLERS WILL BE REQUIRED TO ENGINEER THE APPROPRIATE NETWORK SLICE(S) ●●

local network functions (NFs) – for latency reasons, for example – while others do not.

There is no one-size-fits-all multi-tool for network slicing. In fact, the ability to engineer network slices depends on an evolving toolbox of versatile enablers in five areas: cloud infrastructure, RAN, core, transport, and operations support systems/business support systems (OSS/BSS). Depending on the scenario, different combinations of enablers will be required to engineer the appropriate network slice(s).

Cloud infrastructure

Cloud infrastructure provides great versatility with multiple enablers. The physical infrastructure can be managed, allowing servers, for example, to be allocated to a network slice. Furthermore, the virtualized infrastructure manager enables a set of infrastructure resources to be shared by making use of traditional virtualization. Container-as-a-service provides a more granular and dynamic approach to sharing resources (by using Kubernetes, for example).

The cloud infrastructure is distributed from central data centers (DCs) to the customer on premises and does not have direct awareness of network slices. However, processes can ensure that a cloud platform can fulfill the requirements associated to slices using identifiers of resources. Network slices will have different needs in terms of isolation, distribution and resource guarantees. The cloud infrastructure can provide highly dedicated resources to slices that need it, while other slices share resources. The resources can be optimally used without unnecessary tradeoffs, controlled through orchestration and policies to satisfy the demands of network slices.

RAN

The 3GPP has defined enablers that can be used within a RAN to select appropriate functions and capabilities (such as policies) for network slices. However, the selection of such functions and definition of capabilities relies on implementation. The most important enabler defined in the 3GPP is that of associating each protocol data unit (PDU) session to a slice identifier known as a single network slice selection assistance information (S-NSSAI) as soon as a UE (user equipment) context is created. The RAN reduces a PDU session into dedicated radio bearers (DRBs), which allows the RAN to associate an S-NSSAI to each DRB and to select NFs and capabilities to serve DRB traffic. As an example of capability, a specific next-generation node B central unit user plane (gNB-CU-UP) – hosting PDCP (Packet Data Convergence Protocol) – may be selected for a given S-NSSAI to fulfill delay and security requirements.

Specific layer 1/layer 2 configurations can be tailored for slice policies. A framework for Radio Resource Management (RRM) policies is used to allocate resources and assign QoS levels per slice. For example, the RRM function may use the Radio Resource Partitioning (RRP) capability to allocate a specific partition to a DRB associated to a slice, according to its requirements. Such partitioning may vary depending on slice requirements. Hard partitioning restricts resource usage to a specific slice; soft partitioning allows resources to be used by any slice when they are not utilized by the slice that is nominally assigned to them; shared resources can also be defined for resources accessible by all slices, on demand.

Further, prioritization between DRB traffic can be achieved by means of QoS policies. Such policies allow for the differentiation of DRB traffic within a slice or between slices when shared resources are used.

While it may seem logical to define RRP for each slice supported at the RAN, this is, in practice, suboptimal. There is a tradeoff in performance between the gain of dedicating resources to specific slice services and the overhead in maintaining numerous resource partitions. The balance is to

keep sufficient RRP to guarantee resource isolation per slice where needed, while not impacting radio performance due to excessive partitioning.

Network slice mobility is also enabled by means of signaling between RAN nodes of slice support per tracking area. A mobility function can also take handover decisions on the basis of slice support at the target RAN to achieve radio efficiency while maintaining service continuity.

Core

Core has several enablers, mainly defined by the 3GPP. These enablers make it possible to define dedicated (or shared) user-plane, control-plane or even data-plane NFs at design time that steer the orchestration of the requested network slice at instantiation time. Further, the enablers make it possible to dynamically decide to use dedicated (or shared) user-plane and control-plane NFs based on policy at attach or session requests.

Decisions at design and instantiation times set the context in which the main parameter S-NSSAI (network slice selection assistance information) and secondary parameter Data Network Name (DNN) together with user identities will steer which dedicated or shared NFs that will be used at attach and session requests.

The user-plane function (UPF) is the most valuable NF to dedicate, not only because of the general independency values (optimal redundancy level and no risk of interruption from other services), but also because it ensures low latency through distributed deployment, which makes it possible for the user-data traffic to stay close to customer premises. A dedicated UPF also ensures that established sessions can survive for a period of time, even when the connection to control-plane NFs is lost.

The control-plane NFs are the second-most valuable NFs to dedicate, starting with the session management function (SMF) and followed by the access and mobility management function (AMF). With dedicated distributed SMF and AMF, it is possible to make changes to established sessions and establish new sessions for a period of time, even if connection to data-plane NFs is lost. The final step is

to distribute data-plane NFs such as unified data management (UDM) and unified data repository (UDR) to achieve full independence.

Transport

Traffic flows from one network slice (or a group of them) should be mapped into transport resources that match the required Service Level Agreement (SLA) for the slice or group of slices. It is important to take the capabilities and capacity of the transport infrastructure into consideration when selecting which enablers to use.

There are multiple enablers in the transport domains to support network slicing use cases. For many use cases, it is sufficient to rely on QoS mechanisms (compare differentiated services) and IP ranges as the entry point. This involves configuring end points in the RAN and core to map the traffic into preexisting transport capabilities. Additional marking, such as IPv6 flow labels, can be used to enable the observability and mapping of individual slices. The transport network can be configured further to map the slices to transport VPNs with traffic-engineered characteristics. Dedicated transport VPNs can be used to satisfy slices with highly specific requirements.

Coordinated management is essential between the RAN/core and the transport domain to ensure the end-to-end (E2E) SLAs, which may include cross-domain orchestration. The transport VPNs can be tuned and weighed by the assurance capabilities of the OSS. Transport and mobile network capabilities should be harmonized to ensure that mobile network capabilities are not compromised by limitations in the transport network.

Operations support systems/business support systems

The enablers within the OSS/BSS area relate to the management of service characteristics specified by the SLAs included in contracts. This results in a requirement for assurance and analytics capabilities based on business policies, SLA fulfillment and operations. As both the specific KPIs and the approach to monitoring them will differ between slices, there will be a growing need for customization.

Orchestration enables the automation of manual tasks. For example, if an enterprise requires services in a city location, orchestration automatically configures all the cell sites that provide coverage in that location, generates the configurations required to meet enterprise service SLAs and automatically provisions the respective cells by applying the configurations. This process can adapt policies and NF selection based on slice load and the number of slices supported to deliver a solution that can react to operational conditions while fulfilling SLAs. Orchestration provides faster service provisioning across all the nodes through the automation of every step, including instantiation and provisioning of all necessary network functions.

As the use cases, deployment models and business models are diversified, it must be possible to customize and repeat the actions of the OSS/BSS layer, which drives the adoption of a model- and intent-driven approach, where templates and policies dictate the actions. These templates reflect the SLAs and are used to orchestrate the deployment of NFs, the system's capabilities, configuration and policies. This is essential for speed and cost-efficiency.

Monetization and the timely delivery of services are vital. These, together with a need for cost-efficiency, drive demand for automation and flexibility across the OSS/BSS layer. Exposure enablers are required to allow customers to influence and monitor the service.

Network slicing categories

The main differences between network slices have to do with the geographical area within which their services can be reached and their specific service

●● NETWORK SLICES CAN BE LOOSELY GROUPED INTO THREE SLICING CATEGORIES: CAMPUS-BASED, WIDE-AREA AND LIMITED-AREA ●●

characteristics, particularly in terms of service coverage and enterprise integration depth. Network slices can be loosely grouped into three slicing categories: campus-based, wide-area and limited-area.

In campus-based scenarios, services are mainly consumed locally. In the case of a slice for smart manufacturing plant services, for example, only the base stations covering the plant need to support the plant slices. Core NFs can be deployed in edge DCs within or tightly integrated with the plant, typically together with RAN functions. Depending on the complexity of the particular scenario, either a standalone or hybrid private 5G network is likely to be the best option in these cases. Though not optimal, a virtual private 5G network would also be a possibility.

In wide-area scenarios, services are consumed in a large part of the network. A typical example of this scenario would be services in the transportation/ automobile sector that require wide-area coverage. From a radio perspective, this requires RAN slicing to be set up across the entire network. From a core network perspective, it may require core NFs to be deployed in strategically placed edge DCs, potentially together with RAN functions, throughout the network to guarantee the characteristics and performance of critical functions. In these scenarios, a virtual private 5G network is the ideal choice.

In limited-area scenarios, services are consumed within a geographically limited area, such as a sports arena or massive event. All base stations within this area need to be configured to support the slice, and there needs to be a DC close by for critical functions for the core network and potentially for the RAN. A virtual private 5G network is a good fit in these scenarios.

Campus-based scenarios

Campus-based services require tightly integrated solutions such as indoor coverage and local edge DCs to support low latency, data isolation and service assurance during normal operation and fault situations.

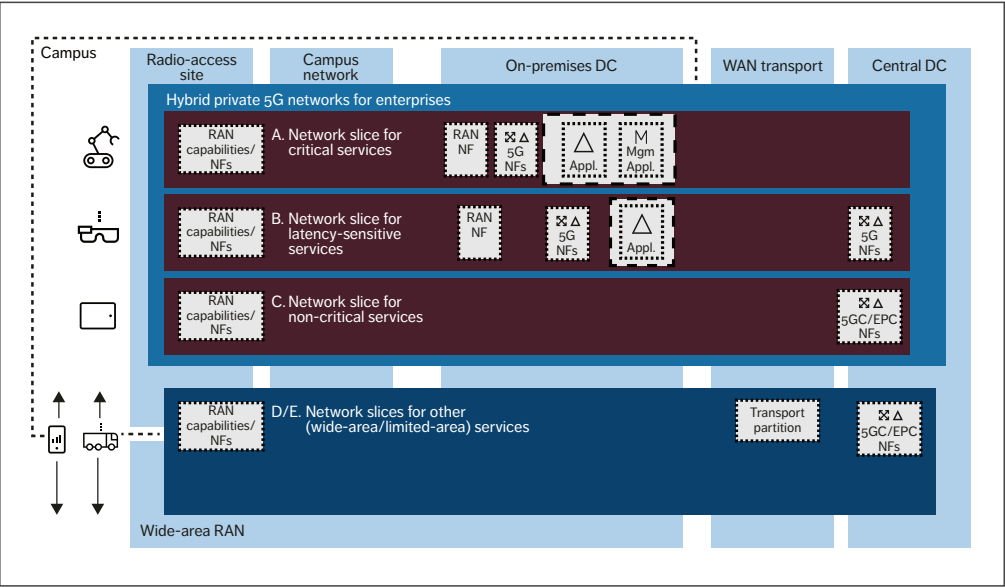


Figure 2 Example of network slicing in a campus deployment (hybrid private 5G network)

Figure 2 illustrates a campus deployment with a hybrid private 5G network that addresses several use cases. A critical slice (A) supports robotics with strong requirements on both latency and survivability. A latency-sensitive slice (B) supports augmented reality/virtual reality application, possibly with lower requirement on survivability. A non-critical slice (C) supports people working at the site (using handheld devices, for example), including mobility across the enterprise sites.

The different types of use cases are supported by one slice each, and each slice is assigned a unique S-NSSAI and DNN. These slices can be assigned dedicated RAN capabilities/NFs to satisfy their unique requirements. For example, services within each slice could use different QoS profiles.

Within the critical service slice (A), hard priority is given (through QoS assignment) to services essential to the smart plant operation. The RAN will then prioritize such essential services in a resource shortage situation. Similarly, RRP in the critical service slice case outlines strict policies for

protecting resource utilization for critical services. To ensure survivability, the critical service slice relies on multiple instantiations of RAN and CN functions to increase redundancy. The network slice for critical services will have a complete on-premises core installation allowing services to survive.

The latency-sensitive slice (B) uses QoS profiles, enabling each service to be served even during high load. This will avoid long latencies and service interruptions, possibly at the expense of throughput per service. Most of the functions handling RAN UP traffic are collapsed into one node, while functions governing control plane traffic are more centralized. The core UPF will be deployed locally to ensure that any delays that occur are minimal.

The non-critical slice (C) can have the complete core installed in the central DC. The RRP covers several different services, whose allocation of resources will be regulated by their QoS profile. Such partitions are provided with "soft" borders, meaning that if resources are not used by the slice services, they are reused by other services.

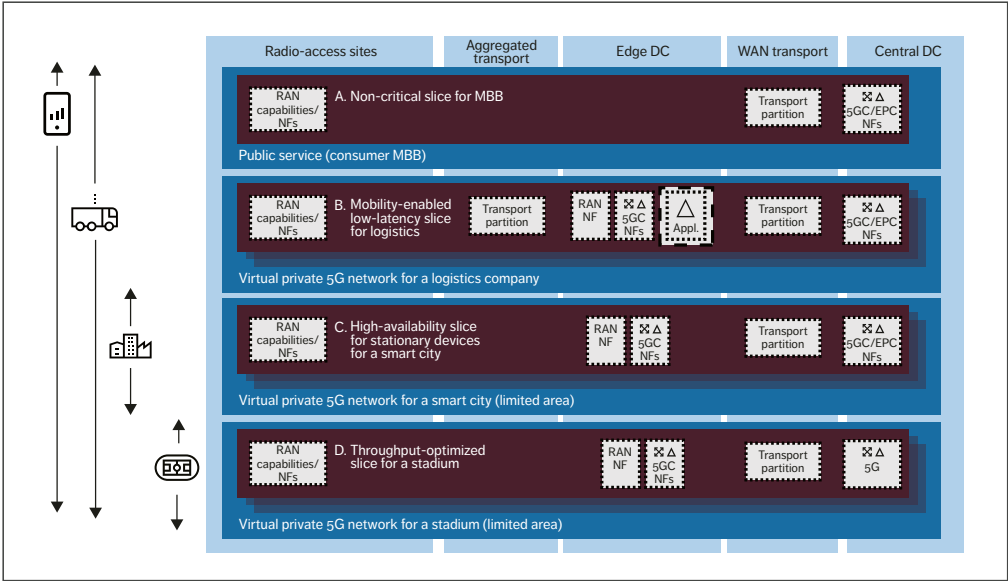


Figure 3 Slicing in wide-area/limited-area deployment (supporting virtual private 5G networks)

As there can be at least as many slices as there are customers, the OSS quickly need to scale to manage a large number of instances. Critical and latency-sensitive traffic would typically stay within the campus, which implies little need to depend on transport slicing for these use cases.

Figure 2 also shows other customers being supported at the site. A slice supporting regular mobile broadband (MBB) service reachable from within the campus (D) allows workers as well as visitors to access the service with full mobility in/out of the campus. Another slice (E) supports virtual private 5G network services for the same enterprise or a different one (such as a logistics company), accessible from within the campus and with mobility in/out of campus.

A specific set of S-NSSAIs and DNNs are used for the MBB slice (D). The core and potentially some RAN functions are deployed in a central DC to provide ubiquitous access. For the wide-area virtual private network (E), a specific set of S-NSSAI and unique DNNs are used to direct to where the

enterprise’s core is installed, either centralized or on-premises. These slices are configured to allow mobility in and out of the campus. To enable this configuration, slice-based mobility features need to be activated, where appropriate target cell selection ensures slice service continuity. The transport network is critical to ensure the right E2E characteristics for these services.

It is preferable to introduce the various enablers in phases, in response to business needs. One approach could be to start with few campuses and then scale up gradually. It is prudent to start with just a few, static slices and then increase the number of slices and the level of “dynamicity” in terms of orchestration and adaptability later on. It is usually wise to focus on less critical or advanced services first, before moving to more critical ones.

Wide-area and limited-area scenarios

The example in Figure 3 shows a combination of wide area network and limited-area use cases. In this context, network slicing enables virtual private 5G

networks. These offerings can be realized as one or several network slices targeting different use cases.

The four use cases shown in Figure 3 – stadium, smart city, logistics company and consumer MBB – each have different requirements in terms of geographical reach and mobility. Network slicing enablers provide each use case with its required functionality and characteristics supporting different levels of SLAs.

A non-critical slice (A) is used for traditional MBB services, providing wide-area service with strong requirements on mobility, interworking with earlier standards, roaming and so on. A mobility-enabled low-latency slice (B) serves the wide-area use case represented by the logistics company in the form of a virtual private 5G network. A high-availability slice (C) is used for stationary devices, serving use cases in a limited area for the smart city use case, as well as in a virtual private 5G network. Finally, a throughput-optimized slice (D) serves use cases in a hotspot for virtual private 5G networks for the sports stadium.

Different use cases and customers may need to be assigned different network slices across the network. The non-critical slice for MBB (A) uses RRP to guarantee bandwidth. Slice-specific QoS regulates resource access among different services. The core network needs to serve full mobility across the coverage but does not require additional distribution.

The mobility-enabled low-latency slice (B) uses RRP available throughout the wide area. It has a dedicated core network with distributed UP and high-availability configuration. Mobility is enabled with robust policies. The high-availability slice for stationary devices (C) uses RRP in a limited area with policies for fairness. The core network is deployed locally without mobility considerations. The throughput-optimized slice (D) also uses RRP in dedicated sites. The core network is optimized for high throughput.

The main challenge for OSS relates to topology. This is due to the fact that even if there are fewer network slices in comparison to the campus case, there may be a large number of sites and areas.

From a phasing perspective, it may make sense to

●● NETWORK SLICING SCENARIOS VARY CONSIDERABLY DUE TO THE DIVERSITY OF USE CASES AND CUSTOMER REQUIREMENTS ●●

start with a few slices with wide-area coverage and a high degree of sharing, and then evolve to more dedicated, isolated and limited-area ones. It is advisable to start with non-critical use cases, and then move on to latency-sensitive ones later.

Conclusion

Network slicing scenarios vary considerably due to the diversity of use cases and customer requirements. Engineering appropriate slices for each case requires a solid understanding of an evolving set of enablers in cloud infrastructure, the RAN, the core and transport networks and in OSS/BSS. In light of this, we at Ericsson believe that the starting point for pursuing network slicing should be the business needs and use cases rather than the technology behind them. To fully harness the power of network slicing, operators will need to embrace a new and transformational approach to building and operating networks.

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Further reading

- » Ericsson, 5G for business, available at: <https://www.ericsson.com/5g/business>
- » Ericsson, Network slicing, available at: <https://www.ericsson.com/network-slicing>
- » Ericsson, 5G RAN slicing, available at: www.ericsson.com/ran-slicing
- » Ericsson, What is 5G?, available at: <https://www.ericsson.com/en/5g/what-is-5g>
- » Ericsson Technology Review, Critical IoT connectivity: Ideal for time-critical communications, June 2, 2020, Fredrik Alriksson, Lisa Boström, Joachim Sachs, Y.-P. Eric Wang, Ali Zaidi, available at: <https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/critical-iot-connectivity>
- » Ericsson white paper, Critical capabilities for private 5G networks, available at: <https://www.ericsson.com/en/reports-and-papers/white-papers/private-5g-networks>
- » GSMA, An Introduction to Network Slicing, available at: <https://www.gsma.com/futurenetworks/wp-content/uploads/2017/11/GSMA-An-Introduction-to-Network-Slicing.pdf>



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